

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 26/10

FULL LENGTH TEST

Solution

English



BPSC TEST-26/10 (SOLUTION)

Q 1. Ans. D

Explanation:

- The Indo-Greeks were the first dynasty to issue coins with bilingual (two languages) and bi-script (two different scripts) legends in India.
- Their coins typically featured the Greek language and script on the obverse (front) along with the portrait of the king, and the Prakrit language written in the Kharoshthi (or sometimes Brahmi) script on the reverse (back). This innovative numismatic practice was a strategic move designed to cater to both the Greek-speaking elites and the local Indian populace, making the currency acceptable across their diverse territories and facilitating smoother trade.
- In contrast, Gupta and Shunga coins predominantly used the Brahmi script, while the Kushanas initially used Greek but soon transitioned to the Bactrian language and script for their coinage.

Q 2. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Harappan civilisation was a Bronze Age culture; they did not have knowledge of iron. To cut and shape hard stones (like carnelian, jasper, and steatite) for bead-making, artisans used specialised drills made of bronze/copper and specialised hard stones (like Ernestite), but no iron tools were used. Iron came into use in the Indian subcontinent much later, during the later Vedic period (around 1000 BCE).
- **Statement 2 is correct:** According to archaeological evidence, Chanhudaro (a tiny settlement compared to Mohenjodaro) was almost exclusively devoted to craft production. It was a highly specialised industrial center known for bead-making, shell-cutting, metal-working, seal-making, and weight-making.

Q 3. Ans. A

Explanation:

- **Statement 1 is correct:** During the Early Vedic (Rig Vedic) period, women enjoyed a respectable and relatively equal position in society. They had access to education and were actively allowed to attend and participate in popular tribal assemblies such as the Sabha and the Vidatha.
- **Statement 2 is correct:** The status of women saw a distinct decline during the Later Vedic period.

Patriarchal norms became more rigid, and texts like the Aitareya Brahmana explicitly referred to the birth of a daughter as a source of misery or sorrow, whereas a son was celebrated as the savior of the family.

- **Statement 3 is incorrect:** Child marriage was **not** prevalent during the Early Vedic period. Historical evidence suggests that girls were married only after attaining maturity/puberty (usually around 16 to 17 years of age). The practices of child marriage and sati did not exist in the Early Vedic society; they emerged in much later periods.

Q 4. Ans. A

Explanation:

- **Assertion (A) is true:** The earliest and most prominent agricultural and urban settlements in the Indian subcontinent, such as the Indus Valley Civilisation (Harappa, Mohenjo-Daro) and the later Vedic settlements in the Gangetic plains, flourished predominantly along major river valleys.
- **Reason (R) is true:** River valleys offered several critical advantages for early human habitation. The rivers deposited nutrient-rich alluvial soil ideal for farming, ensured a steady, year-round water supply for both drinking and agricultural irrigation, and acted as natural highways for trade, transport, and cultural exchange (communication) before the development of extensive road networks.
- **Correct Explanation:** Because rivers provided these indispensable resources—water for crops and navigable routes for movement—they were the direct cause for early agrarian societies choosing to settle and expand in these specific geographic zones. Therefore, Reason (R) perfectly explains Assertion (A).

Q 5. Ans. C

Explanation:

- **Vishti (1):** This refers to forced labour (corvée labour) extracted by the state, king, or local feudal lords from the common people, often without any remuneration. It was considered a form of taxation paid through physical labour.
- **Bhaga (2):** This was the standard land revenue or the king's customary share of the agricultural produce. It was traditionally fixed at one-sixth (shadbhaga) of the total yield.
- **Hiranya (3):** The literal meaning of Hiranya is gold. In the taxation context, it referred to the king's

share of certain crops or taxes that had to be paid in cash (coins/gold) rather than in kind.

- **Udranga (4):** Mentioned frequently in Gupta land grant inscriptions, Udranga was a type of land tax levied specifically on the permanent tenants or settled cultivators of a village. (Another term, Uparikara, was often levied on temporary or cultivating tenants).

Q 6. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Samudragupta issued the Ashvamedha-type gold coins to commemorate his successful performance of the Vedic horse sacrifice, symbolising his imperial conquests. The obverse depicts a horse tied to a sacrificial post (yupa).
- **Pair 2 is correctly matched:** The Archer type was the most widely circulated coin of the Gupta dynasty. While introduced by Samudragupta, it was extensively minted and popularised by **Chandragupta II**. It depicts the king holding a bow in one hand and an arrow in the other.
- **Pair 3 is correctly matched:** The Rhinoceros-slayer type is a unique and famous coin type issued by **Kumaragupta I**. It depicts the king riding a horse and attacking a rhinoceros with a sword. (He also issued other distinct types, like the Kartikeya type).

Q 7. Ans. C

Explanation:

- **Harappan bead making (c. 2500 – 1900 BCE):** This belongs to the Chalcolithic and Bronze Age phase of the Indus Valley Civilisation, representing the earliest urban socio-economic development on this list.
- **Rig Vedic assemblies (c. 1500 – 1000 BCE):** Following the decline of the Harappan civilisation, the early Vedic Aryans settled in the Sapta Sindhu region, where tribal assemblies like the Sabha, Samiti, and Vidatha governed their semi-nomadic, pastoral society.
- **Punch-marked coins (c. 6th Century BCE):** With the Second Urbanisation during the Mahajanapada period (around the time of Buddha), trade expanded significantly, leading to the introduction of India's earliest known coinage, silver and copper punch-marked coins.
- **Post-Gupta land grants (c. 6th Century CE onwards):** Following the decline of the Gupta Empire, the practice of granting land (agrahara/ brahmadeya) to Brahmanas and later to secular officials became widespread, marking the beginning of early medieval Indian feudalism.

Q 8. Ans. C

Explanation:

- **Kailasanatha Temple (3. Kanchipuram):** Built in the 8th century by the Pallava king Rajasimha (Narasimhavarman II), this is one of the oldest structural temples in South India and a prime example of Dravidian architecture.
- **Durga Temple (4. Aihole):** Built by the Chalukyas between the 7th and 8th centuries, this temple is famous for its unique apsidal (horseshoe) plan, which was likely derived from Buddhist chaitya halls.
- **Pancha Rathas (2. Mahabalipuram):** These are magnificent monolithic rock-cut shrines carved out of single granite rock during the reign of Pallava king Narasimhavarman I. They are named after the Pandavas and Draupadi.
- **Papanatha Temple (1. Pattadakal):** Built around 680 CE by the Badami Chalukyas, this temple is a fascinating mix of Northern (Nagara) and Southern (Dravida) architectural styles, characteristic of the Pattadakal monument complex.

Q 9. Ans. A

Explanation:

- **Statement 1 is correct:** The Kailasa temple (Cave 16) at Ellora is one of the most remarkable cave temples in the world because it is not a constructed building, but rather a colossal monolith carved entirely out of a single living rock face. The excavation was done vertically from the top downward.
- **Statement 2 is incorrect:** It was constructed under the patronage of the Rashtrakuta king **Krishna I** (reigned c. 756–773 CE), not Amoghavarsha. Amoghavarsha was a later, very famous Rashtrakuta king, but he was not the patron of the Kailasa temple.

Q 10. Ans. C

Explanation:

- **Statement 1 is correct:** Firoz Shah Tughlaq had a deep fascination with acquiring slaves. During his reign, the number of royal slaves (bandagan) reached an estimated 180,000. To manage this massive workforce, he established a separate and dedicated central department known as the **Diwan-i-Bandagan** (Department of Slaves).
- **Statement 2 is incorrect:** The state did not prohibit their employment in royal workshops; in fact, it was the exact opposite. A significant portion of these slaves was trained in various crafts and arts. Around 12,000 of them were actively employed in the royal

karkhanas (state workshops) to manufacture goods, weapons, and textiles for the royal household and the military.

- **Statement 3 is correct:** Firoz Shah took care of the upkeep of his slaves at the expense of the state treasury. According to contemporary historian Shams-i-Siraj Afif, these state-owned slaves were given fixed cash salaries ranging from 10 to 100 tankas per annum, and some were even granted land assignments (iqtas) in lieu of cash.

Q 11. Ans. C

Explanation:

- **Lala Lajpat Rai** prominently popularised the “Safety Valve” theory in his 1916 book, *Young India*.
- According to this theory, the Indian National Congress was founded by the retired British civil servant A.O. Hume under the secret guidance (or as a product of the “brain”) of the then Viceroy, Lord Dufferin.
- Extremist leaders like Lala Lajpat Rai used this theory to attack the Moderates, arguing that the Congress was primarily established as a ‘safety valve’ to peacefully release the growing popular discontent and avert a massive anti-British uprising similar to the Revolt of 1857.
- By framing the INC’s origins this way, Rai highlighted that the organisation was initially designed to protect the British Empire rather than to genuinely champion Indian political independence.

Q 12. Ans. A

Explanation:

- **Statement 1 is correct:** Warren Hastings initiated the “Ring Fence” policy, which aimed to create buffer zones to protect the British East India Company’s frontiers. Instead of directly defending their own borders, the British defended the borders of neighbouring states. For example, they allied with Awadh to protect it from Maratha and Afghan invasions, which in turn served as a buffer to protect the Company’s core territory of Bengal.
- **Statement 2 is incorrect:** Lord Dalhousie annexed the Kingdom of Awadh in 1856 strictly on the pretext of **misgovernance** (maladministration), not under the Doctrine of Lapse. The Nawab of Awadh, Wajid Ali Shah, had plenty of legitimate heirs, so the Doctrine of Lapse, which was used to annex states lacking a natural male heir, could not be applied. States that were actually annexed under the Doctrine of Lapse included Satara, Jhansi, Nagpur, and Sambalpur.

Q 13. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Arya Samaj, founded by Swami Dayananda Saraswati, regarded the **Vedas** as the supreme, infallible authority, not the Upanishads. His foundational slogan for the movement was “Go back to the Vedas,” as he believed they contained the ultimate truth and all scientific knowledge, free from later corruptions.
- **Statement 2 is correct:** The Prarthana Samaj (founded by Dr. Atmaram Pandurang and championed by M.G. Ranade) did not focus on creating a new theology or breaking entirely from Hinduism. Instead, it was fundamentally a social reform movement. Its primary agenda included the abolition of the caste system, introducing widow remarriage, encouraging female education, and raising the age of marriage for both males and females.
- **Statement 3 is correct:** The Ramakrishna Mission, founded by Swami Vivekananda, was built on two core pillars. The first was the teaching of his guru, Ramakrishna Paramahansa, which stressed the essential spiritual unity of all religions (that all religions are true paths leading to the same divine reality). The second pillar was Vivekananda’s doctrine of humanitarian service, captured in his belief that “service to man is service to God” (Daridra Narayana).

Q 14. Ans. A

Explanation:

- **Assertion (A) is true:** After losing the fort of Chunar to Mughal Emperor Humayun, Sher Shah Suri acquired the highly strategic hill fort of Rohtasgarh in Bihar in 1539. He extensively fortified its defences with thicker stone walls and bastions adapted for artillery to make it an impenetrable stronghold.
- **Reason (R) is true:** The primary motivation for capturing and fortifying this specific fort was strategic necessity. With Humayun advancing and Sher Shah engaged in campaigns in Bengal (Gauda), he desperately needed a secure sanctuary. He famously used a “Trojan Horse” ruse, asking the local Hindu king of Rohtas to provide a safe haven for his women, children, and the immense treasure he had looted from Bengal, only to smuggle in fierce Afghan soldiers in the palanquins to capture the fort.
- **Correct Explanation:** Because he required an impregnable base to safeguard his family and his newly acquired wealth while he was away fighting, he targeted and reinforced Rohtasgarh. Therefore, (R) is the direct and correct explanation for (A).

Q 15. Ans. A

Explanation:

- **Regulating Act of 1773 (3):** This was the first major step by the British Parliament to control and regulate the affairs of the East India Company. A key provision was the establishment of a Supreme Court at Calcutta (set up in 1774) comprising one Chief Justice and three other judges.
- **Charter Act of 1813 (1):** This Act abolished the East India Company's monopoly over Indian trade, opening the market to all British merchants. However, the Company retained its exclusive monopoly over the tea trade and trade with China for another 20 years.
- **Charter Act of 1833 (4):** This Act was the final step towards complete centralisation in British India. It elevated the "Governor-General of Bengal" to the "Governor-General of India" (making Lord William Bentinck the first to hold the title) and vested in him all civil and military powers across British India.
- **Government of India Act 1858 (2):** Enacted immediately following the Revolt of 1857, this Act liquidated the East India Company and transferred the powers of government, territories, and revenues directly to the British Crown. It created the new office of the **Secretary of State for India**, who was a member of the British Cabinet and exercised absolute authority over Indian administration.

Q 16. Ans. D

Explanation:

- **Pair 1 is correctly matched: Dadabhai Naoroji** (the "Grand Old Man of India") authored the seminal book *Poverty and Un-British Rule in India* (1901), where he extensively formulated the "Drain of Wealth" theory, arguing that a portion of India's wealth and resources was being continuously exported to Britain without any adequate economic, commercial, or material return.
- **Pair 2 is correctly matched: Romesh Chunder Dutt (R.C. Dutt)**, a retired ICS officer, wrote the monumental two-volume work *The Economic History of India* (1902–1904). It was the most detailed and analytical economic critique of the time, focusing heavily on British land revenue policies and the deindustrialisation of Indian handicrafts.
- **Pair 3 is correctly matched: Mahadev Govind Ranade (M.G. Ranade)**, a prominent Maharashtrian reformer, published *Essays on Indian Economics* (1898). He argued against applying classical

English free-trade economics to India, advocating instead for state-sponsored industrialisation and protection of nascent Indian industries.

Q 17. Ans. C

Explanation:

- The decisive Battle of Wandiwash (fought in 1760) was a part of the **Third Carnatic War** (1756–1763), which was the Indian theatre of the global Seven Years' War happening in Europe.
- In this battle, the British forces under Sir Eyre Coote decisively defeated the French forces led by the Comte de Lally. This victory severely crippled the French military presence in India, leading to the capture of their capital, Pondicherry, a year later. The Treaty of Paris (1763) later restored French factories, but strictly confined them to commercial activities, permanently ending French imperial and political ambitions in the Indian subcontinent.

Q 18. Ans. C

Explanation:

- **Statement 1 is incorrect:** The Anarchical and Revolutionary Crimes Act of 1919, popularly known as the Rowlatt Act, was enacted based on the recommendations of the Sedition Committee chaired by Sir Sidney Rowlatt. It gave the British colonial government enormous powers to repress political activities and allowed the detention of political prisoners without trial for up to two years (summarised by the popular slogan "No Dalil, No Vakil, No Appeal").
- **Statement 2 is correct:** In response to the massive outrage over the Jallianwala Bagh massacre (April 1919), the government formed the Disorders Inquiry Committee, widely known as the Hunter Committee (or Commission). Out of its members, three were distinguished Indians: Sir Chimanlal Harilal Setalvad, Pandit Jagat Narayan, and Sardar Sahibzada Sultan Ahmad Khan. Their inclusion was intended to give the committee a semblance of impartiality, though the final report still deeply disappointed Indian nationalists.

Q 19. Ans. A

Explanation:

- **Statement 1 is correct:** "Home Charges" formed a massive and crucial component of the Drain of Wealth. These were expenses incurred in England but paid from Indian tax revenues. They included the pensions and salaries of British civil and military officials on furlough, interest on the East

India Company's public debt, dividends paid to its shareholders, and the cost of maintaining the Secretary of State's office in London.

- **Statement 2 is correct:** After acquiring the Diwani (revenue rights) of Bengal in 1765, the East India Company stopped bringing gold and bullion from Britain to buy Indian commodities. Instead, they began using the tax revenues forcibly collected from Indian peasants to purchase Indian goods (officially termed as 'investments'). These goods were then exported to Europe for profit, meaning India was effectively paying for its own exports without receiving anything in return.
- **Statement 3 is incorrect:** Early nationalist economic thinkers (like Dadabhai Naoroji, R.C. Dutt, and M.G. Ranade) were profoundly suspicious and highly critical of the influx of foreign capital. They argued that foreign investment, particularly in railways and plantations, suppressed indigenous enterprise and did not lead to genuine industrialisation. Instead, it drastically increased the drain of wealth through the continuous repatriation of massive profits, dividends, and guaranteed interest back to Britain.

Q 20. Ans. A

Explanation:

- **Assertion (A) is true:** The Pitt's India Act of 1784 is historically significant for establishing a system of "double government" in the administration of British India.
- **Reason (R) is true:** For the first time, the Act explicitly distinguished between the commercial and political functions of the East India Company.
- **Correct Explanation:** To achieve this separation, the Act allowed the existing Court of Directors to continue managing the Company's commercial trade. However, it created a new, supreme administrative body called the Board of Control (which reported directly to the British Parliament) to manage and direct all political, civil, and military affairs. Because this separation of functions created two distinct and parallel governing authorities, it established the system of double government. Therefore, (R) is the correct explanation for (A).

Q 21. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The First Carnatic War (1746–1748) was an extension of the War of the Austrian Succession in Europe. It concluded with the signing of the **Treaty of Aix-la-Chapelle**

in 1748. Under this treaty, Madras was restored to the English in exchange for the French gaining Louisbourg in North America.

- **Pair 2 is correctly matched:** The Second Carnatic War (1749–1754) was concluded with the **Treaty of Pondicherry** (1754). It was signed by Charles Godeheu, who had replaced the ambitious Joseph François Dupleix as the French Governor-General. Under its terms, both the English and the French agreed not to interfere in the internal political quarrels of native Indian princes.
- **Pair 3 is incorrectly matched:** The Third Anglo-Mysore War (1790–1792) culminated in the **Treaty of Seringapatam** (1792), signed by Lord Cornwallis and Tipu Sultan (which forced Tipu to surrender half of his kingdom). The **Treaty of Paris** (1763), on the other hand, was the agreement that ended the Third Carnatic War (part of the global Seven Years' War).

Q 22. Ans. C

Explanation:

- The All India Kisan Sabha (AIKS) was founded in April 1936 at the Lucknow session of the Indian National Congress. Its objective was to mobilise peasants against the oppressive zamindari system and demand agrarian reforms.
- **Swami Sahajanand Saraswati**, a brilliant intellectual and a formidable peasant leader from Bihar, was elected as its first President. N.G. Ranga, the pioneer of the peasant movement in Andhra, was elected as its first General Secretary.

Q 23. Ans. A

Explanation:

- **Statement 1 is correct:** The Congress Socialist Party (CSP) was founded in 1934 by socialist-leaning members of the Indian National Congress. It was not a separate political party but rather a left-wing caucus operating entirely within the Congress. Its founders wanted to shape the Congress's ideology towards socialism and integrate peasant and worker struggles into the broader national movement without breaking away from the main anti-imperialist organisation.
- **Statement 2 is incorrect:** **Acharya Narendra Dev** served as the first President of the Congress Socialist Party, while **Jayaprakash Narayan** was its first General Secretary.

Q 24. Ans. A

Explanation:

- **Statement 1 is correct:** The All-India Muslim

League formally adopted the Lahore Resolution (later known as the Pakistan Resolution) during its annual session on March 23-24, 1940. The resolution formally demanded that the Muslim-majority areas in the north-western and eastern zones of India be grouped to constitute "Independent States" in which the constituent units would be autonomous and sovereign.

- **Statement 2 is correct:** The original draft of the resolution was primarily authored by **Sir Sikandar Hayat Khan**, the Premier of Punjab. It was later presented in the session by A.K. Fazlul Huq, the Premier of Bengal, and seconded by Choudhry Khaliquzzaman.
- **Statement 3 is incorrect:** Interestingly, the original text of the Lahore Resolution **did not mention the word "Pakistan"** anywhere. It only spoke of "Independent States." It was the Hindu and British press that quickly dubbed it the "Pakistan Resolution." The Muslim League leadership eventually embraced this name. (The term "Pakistan" itself had been coined earlier in 1933 by a student named Choudhry Rahmat Ali in his pamphlet *Now or Never*)

Q 25. Ans. A

Explanation:

- **Assertion (A) is true:** During the famous Red Fort trials of 1945-46, three prominent Indian National Army (INA) commanders, Prem Kumar Sahgal, Shah Nawaz Khan, and Gurbaksh Singh Dhillon, were found guilty of "waging war against the King-Emperor." However, under immense pressure, the Commander-in-Chief of the British Indian Army, General Claude Auchinleck, eventually remitted their sentences and released them.
- **Reason (R) is true:** The INA trials acted as a lightning rod for Indian nationalism. They triggered unprecedented nationwide mass upsurges, strikes, and protests. Most alarmingly for the British Empire, this nationalist fervour heavily infected the regular armed forces, culminating in events like the Royal Indian Navy (RIN) Mutiny of 1946 and widespread unrest in the Air Force and Army.
- **Correct Explanation:** The British colonial government did not remit the sentences out of goodwill. General Auchinleck was forced to abandon the punishments specifically because he realised that enforcing them would lead to complete chaos and a full-scale revolt within the Indian ranks of the British military. Thus, the massive public upsurges directly dictated the government's retreat, making (R) the perfect and correct explanation for (A).

Q 26. Ans. B

Explanation:

- **Rani Jhansi Regiment (3. Lakshmi Sahgal):** This was the all-women combat regiment of the Indian National Army (INA), raised in 1943. It was commanded by Captain Lakshmi Swaminathan (later known as Lakshmi Sahgal) and was one of the very few all-female combat regiments in the world at the time.
- **Free India Centre (4. Subhas Chandra Bose):** Founded by Subhas Chandra Bose in Berlin, Germany in 1941. It managed his political and diplomatic activities in Europe, published the *Azad Hind* magazine, and was the organisation that first coined the famous national greeting "Jai Hind."
- **Indian Independence League (1. Rash Behari Bose):** Established by the veteran revolutionary Rash Behari Bose in 1942 in Southeast Asia. Its purpose was to organise the Indian diaspora to fight for India's independence. He later handed over the leadership of the League and the INA to Subhas Chandra Bose in 1943.
- **INA Defence Committee (2. Bhulabhai Desai):** Formed by the Indian National Congress in 1945 to defend the INA officers (Sahgal, Dhillon, and Shah Nawaz Khan) facing treason charges at the historic Red Fort trials. The legal defence team was famously led by the eminent lawyer Bhulabhai Desai, and included other legal luminaries like Tej Bahadur Sapru, Kailash Nath Katju, and Jawaharlal Nehru.

Q 27. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The **All India Trade Union Congress (AITUC)** was founded in 1920 to represent India's rapidly growing labour movement. The prominent nationalist leader **Lala Lajpat Rai** was elected as its first President, and Dewan Chaman Lall served as its first General Secretary.
- **Pair 2 is incorrectly matched:** The **Independent Labour Party** was founded by **Dr. B.R. Ambedkar** in 1936 to represent the interests of the depressed classes and the working poor against both capitalist and Brahmanical structures. N.M. Joshi, on the other hand, was a pioneer of the trade union movement who founded the Social Service League and later the All India Trade Union Federation (AITUF).
- **Pair 3 is correctly matched:** The **Bihar Provincial Kisan Sabha** was established in 1929 by **Swami Sahajanand Saraswati**. He mobilised the peasantry against the zamindari system and the oppressive

attacks on their occupancy rights. The massive success of this provincial movement eventually laid the groundwork for the formation of the All India Kisan Sabha in 1936.

Q 28. Ans. A

Explanation:

- **2. The Labour Government came to power in Britain (July 1945):** Following the end of World War II in Europe, general elections were held in the United Kingdom. On **26 July 1945**, the Labour Party, led by Clement Attlee, won a decisive victory, replacing Winston Churchill's Conservative government. This marked a significant shift toward a more responsive British policy regarding Indian independence.
- **1. Elections to the Central and Provincial Legislative Assemblies were announced (August 1945):** Immediately after assuming power, the new Labour government summoned Viceroy Lord Wavell to London for consultations. Upon his return, Wavell announced on **21 August 1945** that general elections to the Central and Provincial Legislative Assemblies would be held in the coming winter of 1945–46 (the previous elections had been held in 1934 and 1937, respectively).
- **4. Announcement regarding the setting up of a Constituent Assembly after the War (September 1945):** In a major broadcast on **19 September 1945** (popularly known as Wavell's September Broadcast), Lord Wavell officially announced the British government's intention to take steps immediately following the assembly elections to convene a constitution-making body (Constituent Assembly) to frame India's new Constitution.
- **3. Strike by the Royal Indian Navy Ratings began in Bombay (February 1946):** On **18 February 1946**, ratings (non-commissioned sailors) of the Royal Indian Navy aboard the HMIS Talwar in Bombay went on strike to protest against racial discrimination, poor food quality, and abuse by British officers. This rebellion quickly spread to shore establishments and over 70 ships across India, becoming a massive anti-imperialist uprising known as the RIN Mutiny.

Q 29. Ans. D

Explanation:

- **2. The Clement Attlee Statement (February 20, 1947):** British Prime Minister Clement Attlee declared in the House of Commons that the British government would grant full self-government to

British India by a strict deadline of June 30, 1948, at the latest. He also announced the appointment of Lord Mountbatten as the new Viceroy to oversee this transition.

- **3. The Mountbatten Plan (June 3, 1947):** Also known as the "June 3 Plan," Lord Mountbatten formally announced this scheme, which laid out the specific mechanics for the partition of British India into two independent dominions: India and Pakistan. Both the Congress and the Muslim League accepted it.
- **1. The Indian Independence Act passed (July 18, 1947):** Based entirely on the Mountbatten Plan, the Indian Independence Bill was introduced in the British Parliament on July 4, 1947, and rapidly received Royal Assent on July 18, 1947, officially becoming an Act.
- **4. The Act came into force (August 15, 1947):** The provisions of the Indian Independence Act came into full legal effect at midnight on August 14–15, 1947. British rule officially ended, and India and Pakistan emerged as newly independent nations.

Q 30. Ans. B

Explanation:

- **Statement 1 is correct:** To secure Indian cooperation during World War II, the Cripps Mission (led by Sir Stafford Cripps) promised to grant India Dominion Status with the explicit freedom to remain in or secede from the British Commonwealth. However, this was strictly a post-war promise, with no immediate constitutional changes offered.
- **Statement 2 is correct:** The proposals contained a highly controversial "opt-out" clause, which allowed any province unwilling to join the new Indian Union to form a separate dominion. Because the proposals offered only future promises rather than immediate independence, Mahatma Gandhi famously dismissed the entire offer as a "post-dated cheque on a crashing/failing bank."
- **Statement 3 is correct:** The failure of the Cripps Mission was largely due to the British refusal to immediately transfer substantial power. The Congress demanded that the Viceroy's Executive Council be immediately reconstituted as a genuine National Government (a cabinet responsible to the Central Legislature) and that the Defence portfolio be handed over to an Indian. Cripps refused both demands, offering only an advisory role.
- **Statement 4 is incorrect:** The Muslim League did not reject the proposals because they failed to guarantee

a single sovereign Pakistan. In fact, the League's objection was precisely the opposite: the opt-out clause for provinces was considered too weak and indirect a path to Pakistan. Jinnah objected that the proposals did not concede Pakistan unambiguously and directly, and that the mechanism for partition was too complicated and uncertain. The League wanted explicit recognition of Pakistan as a sovereign state, not merely a provincial opt-out provision. Crucially, it was also dissatisfied that the opt-out clause applied to provinces individually rather than to pre-defined Muslim-majority zones as a bloc.

Q 31. Ans. B

Explanation:

- **Statement 1 is incorrect:** Viruses are **acellular infectious agents** that contain **either DNA or RNA as their genetic material, but never both simultaneously**. The nucleic acid is enclosed within a protective protein coat called a **capsid**. Depending on the type of virus, the genetic material may be DNA (DNA viruses) or RNA (RNA viruses). Therefore, the statement that viruses possess both DNA and RNA is incorrect.
- **Statement 2 is correct:** viruses are **obligate intracellular parasites**, meaning they can reproduce only inside the living cells of a host organism. Outside a living cell, viruses remain metabolically inactive and cannot carry out replication because they lack the cellular machinery required for protein synthesis and reproduction. They use the host cell's enzymes and ribosomes to produce new virus particles. Therefore, the statement is correct.

Q 32. Ans. B

Explanation:

- Hydra belongs to the phylum **Cnidaria** (Coelenterata). Members of this phylum are **aquatic, mostly marine** (though Hydra is freshwater), exhibit **radial symmetry**, and possess **specialised stinging cells called cnidocytes (cnidoblasts)**, which are used for capturing prey and defence. Hydra has a simple body with a single opening (mouth) serving for both ingestion and egestion, making it a characteristic cnidarian.

Q 33. Ans. C

Explanation:

- **Pair 1 is correct:** **Agrostology** is the branch of botany that deals with the **scientific study of grasses (family Poaceae)**. Grasses are economically important as

they include cereals such as rice, wheat, and maize. This pair is correctly matched.

- **Pair 2 is incorrect:** **Oncology** is the branch of medical science concerned with the **study, diagnosis, and treatment of cancer**. The study of **ageing** is known as **Gerontology**, not Oncology. Therefore, this pair is incorrectly matched.
- **Pair 3 is correct:** **Malacology** is the branch of zoology that deals with the **study of molluscs**, including snails, oysters, clams, squids, and octopuses. This pair is correctly matched.
- **Pair 4 is correct:** **Phrenology** is the historical study of the **shape and contours of the skull**, based on the old belief that skull shape reflected mental faculties and personality traits. Although **phrenology is now regarded as a pseudoscience**, the term itself correctly refers to the study of skull shape. Hence, this pair is correctly matched.

Q 34. Ans. C

Explanation:

- The term "**protoplasm**" was first coined by **J. E. Purkinje (Jan Evangelista Purkyně)** in **1839** to describe the **living substance of the cell**. Protoplasm comprises all the living contents enclosed by the plasma membrane, including the **cytoplasm and the nucleus**. It serves as the site for almost all vital cellular activities such as metabolism, growth, respiration, protein synthesis, and cell division. Because it contains the organelles and genetic material essential for life processes, protoplasm was historically regarded as the "**physical basis of life**." Therefore, Purkinje is correctly credited with introducing the term **protoplasm**.

Q 35. Ans. A

Explanation:

- **Statement 1 is correct:** Some species of the fungus **Aspergillus** produce **aflatoxins**, which are highly toxic secondary metabolites. *Aspergillus flavus* commonly contaminates food crops such as **maize, groundnuts (peanuts), tree nuts, and other stored grains**, especially under warm and humid conditions. Aflatoxins are **carcinogenic toxins** that contaminate food and pose serious health hazards to humans and animals. Therefore, this statement is correct.
- **Statement 2 is incorrect:** Aflatoxins are **toxic compounds produced by fungi**, not substances that inhibit fungal growth. They do **not** function as antifungal agents or prevent mould formation. Instead, they are harmful contaminants that can

cause liver damage and increase the risk of cancer when contaminated food is consumed. Hence, the statement is incorrect.

Q 36. Ans. A

Explanation:

- **Messenger RNA (mRNA)** is synthesised inside the **nucleus** during the process of **transcription**, in which genetic information stored in DNA is copied into an RNA molecule. After its synthesis and processing, mRNA exits the nucleus through the **nuclear pores** and enters the **cytoplasm**. In the cytoplasm, it binds to **ribosomes** and acts as a **template for protein synthesis (translation)** by carrying the genetic code in the form of codons. Since mRNA is present in the **nucleus during its formation** and in the **cytoplasm during protein synthesis**, it is the only component among the given options that is normally found in **both the nucleus and the cytoplasm** of a eukaryotic cell. Hence, **mRNA is the correct answer**.

Q 37. Ans. D

Explanation:

- **Leprosy** is a chronic infectious disease caused by *Mycobacterium leprae*. The bacterium primarily affects the **peripheral nervous system**, damaging peripheral nerves and causing loss of sensation, muscle weakness, and skin lesions. If left untreated, it can lead to permanent nerve damage and deformities. Therefore, the nervous system is correctly matched with **Leprosy**.
- **Pneumonia** is a bacterial or viral disease affecting the **lungs**, which are part of the respiratory system. The alveoli become inflamed and may fill with fluid or pus, leading to symptoms such as cough, fever, chest pain, and difficulty in breathing. Since pneumonia primarily affects the respiratory tract, this pair is correctly matched.
- **Gonorrhoea** is a **sexually transmitted bacterial disease** caused by *Neisseria gonorrhoeae*. It infects the **genital and urinary (genitourinary) tracts**, producing inflammation and painful urination. Because the disease commonly affects the **urinary tract** along with the reproductive tract, this pair is correctly matched.
- **Syphilis** is a **sexually transmitted disease (STD)** caused by *Treponema pallidum*. The infection is transmitted through sexual contact and initially affects the **reproductive tract**, producing sores and lesions before spreading to other parts of the body if untreated. Therefore, the reproductive tract is correctly matched with **Syphilis**.

Q 38. Ans. D

Explanation:

- **Statement:** A **species** is the **basic or fundamental unit of classification**. A species is defined as a group of organisms that share similar morphological and genetic characteristics and are capable of interbreeding under natural conditions to produce fertile offspring. The taxonomic hierarchy is arranged as **Species → Genus → Family → Order → Class → Phylum/Division → Kingdom**, with species forming the lowest and most specific category.
- **Conclusion I:** Organisms classified under the **same species** exhibit the **maximum degree of similarity** in their morphology, genetics, physiology, and reproductive characteristics. As we move higher in the taxonomic hierarchy—from species to genus, family, order, and beyond—the number of shared characteristics progressively decreases. Therefore, organisms belonging to the same species are much more closely related than organisms belonging to different genera. Hence, **Conclusion I is correct**.
- **Conclusion II:** **Species is the lowest and most specific taxonomic category**, not the highest. The **highest taxonomic category** in the NCERT hierarchy is **Kingdom**. Each higher category includes one or more lower categories and represents a broader group of organisms with fewer common characteristics. Therefore, **Conclusion II is incorrect**.

Q 39. Ans. C

Explanation:

- The image formed by a convex lens is determined using the lens formula:
 - ✓ $1/f = 1/v - 1/u$
 - ✓ For a convex lens:
 - ✓ Focal length (f) = +20 cm
 - ✓ Object distance (u) = -30 cm (as per the Cartesian sign convention)
 - ✓ Substituting the values:
 - ✓ $1/20 = 1/v + 1/30$
 - ✓ $1/v = 1/20 - 1/30 = (3 - 2)/60 = 1/60$
 - ✓ Therefore, $v = +60$ cm
- The positive sign indicates that the image is formed on the opposite side of the lens. Since the object is placed between the principal focus ($F = 20$ cm) and twice the focal length ($2F = 40$ cm), the **image is formed beyond $2F$, and it is real, inverted, and magnified**.

Q 40. Ans. B**Explanation:**

- When a ray of light passes from air (rarer medium) into glass (denser medium), it bends towards the normal. The angle of refraction is calculated using Snell's Law:
 - $n = \sin i / \sin r$
 - Where:
 - ✓ Refractive index of glass (n) = 1.5
 - ✓ Angle of incidence (i) = 45°
- Substituting the values:
- $1.5 = \sin 45^\circ / \sin r$
 - $\sin r = \sin 45^\circ / 1.5$
 - $\sin r = 0.707 / 1.5 = 0.471$
 - $r = \sin^{-1}(0.471) \approx 28^\circ$
 - Thus, the angle of refraction is approximately 28° . Since light enters a denser medium, it bends towards the normal, making the angle of refraction smaller than the angle of incidence. Therefore, the correct answer is 28° .

Q 41. Ans. B**Explanation:**

- The weight of an object is the gravitational force exerted by the Earth on it and is given by:
- Weight (W) = Mass (m) $\times$ Acceleration due to gravity (g)
- The value of g is not the same everywhere on the Earth's surface. It is maximum at the poles and minimum at the equator because:
 - ✓ The Earth is slightly flattened at the poles and bulges at the equator. Therefore, the distance from the Earth's centre to the poles is slightly less than that to the equator, resulting in a stronger gravitational pull at the poles.
 - ✓ The Earth's rotation produces a centrifugal force, which is maximum at the equator and zero at the poles. This centrifugal force opposes gravity and reduces the effective value of g at the equator.
- At the centre of the Earth, the gravitational force becomes zero because the gravitational pull from all directions balances out. Hence, an object has zero weight at the Earth's centre.

Q 42. Ans. D**Explanation:**

- Capillary rise is the phenomenon in which a liquid rises or falls in a narrow tube due to the combined effect of adhesive forces, cohesive forces, and surface tension.

- In the case of water in a glass capillary tube, the adhesive force between water molecules and the glass surface is greater than the cohesive force between water molecules. As a result, water wets the glass and forms a concave meniscus. The surface tension acting along the curved surface of the liquid pulls the water upward until the upward force due to surface tension is balanced by the weight of the liquid column.
- Thus, the rise of water in a capillary tube occurs because of the combined action of adhesive and cohesive forces, while surface tension transmits the effect of these intermolecular forces. Therefore, the correct answer is Adhesive and cohesive forces.

Q 43. Ans. C**Explanation:**

- **Statement:** Insulin is a peptide hormone secreted by the β -cells of the Islets of Langerhans in the pancreas. It plays a central role in maintaining normal blood glucose levels by promoting the uptake, utilisation, and storage of glucose in body cells. Through these actions, insulin helps maintain glucose homeostasis in the body.
- **Conclusion I:** Insulin lowers blood glucose concentration by facilitating the uptake of glucose into body cells, particularly muscle and adipose tissue. It also stimulates the conversion of excess glucose into glycogen in the liver and muscles (glycogenesis). By increasing glucose utilisation and storage, insulin effectively reduces the level of glucose circulating in the blood. Therefore, Conclusion I is correct.
- **Conclusion II:** The endocrine portion of the pancreas contains clusters of cells known as the Islets of Langerhans. Among these, the β -cells (beta cells) are responsible for the secretion of insulin, while the α -cells (alpha cells) secrete glucagon. Thus, insulin is specifically produced and released by the β -cells of the pancreas, making Conclusion II correct.

Q 44. Ans. C**Explanation:**

- Most solids expand when heated and contract when cooled. The pendulum rod of a pendulum clock is usually made of a metal, which undergoes linear expansion in hot weather. As a result, the length of the pendulum increases.
- The time period of a simple pendulum is given by:
 - ✓ $T = 2\pi \sqrt{L/g}$

- As the length (L) of the pendulum increases, its **time period also increases**. This means each oscillation takes a little more time to complete, causing the pendulum to swing more slowly. Consequently, the clock **loses time**, or in other words, **slows down** in hot weather.
- In cold weather, the pendulum contracts, its length decreases, the time period becomes shorter, and the clock tends to run faster. Therefore, the correct answer is: Clock slows down in hot weather.

Q 45. Ans. C

Explanation:

- **Statement:** Enzymes are **biological catalysts** that accelerate biochemical reactions occurring in living cells without being consumed in the process. They increase the rate of reactions by providing an alternative pathway with a lower energy requirement, thereby enabling metabolic reactions to occur efficiently under normal physiological conditions.
- **Conclusion I:** Enzymes speed up biochemical reactions by **lowering the activation energy**, which is the minimum energy required to initiate a chemical reaction. They provide an alternative reaction pathway with a lower energy barrier, allowing more reactant molecules to be converted into products in a given time. Thus, lowering activation energy is the fundamental reason why enzymes act as effective biological catalysts. Therefore, **Conclusion I is correct**.
- **Conclusion II:** Enzymes participate in the reaction by temporarily forming an **enzyme-substrate complex**, but after the reaction is completed, the products are released, and the enzyme regains its original structure. Since enzymes are **not consumed or permanently altered during the reaction**, the same enzyme molecule can catalyse the reaction repeatedly. Therefore, **Conclusion II is correct**.

Q 46. Ans. D

Explanation:

- **Statement 1 is correct:** a liquid boils when its **vapour pressure becomes equal to the external pressure**. If the external pressure is increased, the liquid must be heated to a higher temperature for its vapour pressure to equal the surrounding pressure. Therefore, the **boiling point increases with an increase in external pressure**. This is why water boils above 100°C in a pressure cooker.
- **Statement 2 is correct:** ice is an exception among

solids because its solid form is **less dense than liquid water**. When pressure is increased, the denser liquid phase is favoured. As a result, the **melting point of ice decreases slightly with increasing pressure**. This phenomenon also explains **regelation**, where ice melts under pressure and refreezes when the pressure is removed.

Q 47. Ans. D

Explanation:

- **Bernoulli's theorem** states that for an **ideal fluid** (incompressible, non-viscous, and flowing steadily), the **total mechanical energy per unit volume** remains constant along a streamline.
- **The theorem is expressed as:**
 - ✓ $P + \frac{1}{2}\rho v^2 + \rho gh = \text{Constant}$
 - ✓ **Where:**
 - P = Pressure energy per unit volume
 - $\frac{1}{2}\rho v^2$ = Kinetic energy per unit volume
 - ρgh = Potential energy per unit volume
- Thus, the total energy per unit volume of a flowing fluid is the **sum of pressure energy, kinetic energy, and potential energy**. If one form of energy increases, one or both of the other forms decrease so that the total energy remains constant. Therefore, the correct answer is **Pressure, kinetic and potential energy**.

Q 48. Ans. C

Explanation:

- **Statement 1 is incorrect:** Thorium-232 is a **fertile material**, not a fissile one. This means it cannot undergo nuclear fission directly to sustain a chain reaction on its own. Instead, it must first absorb a neutron to transmute (convert) into Uranium-233, which is a fissile material capable of sustaining a chain reaction.
- **Statement 2 is correct:** India possesses one of the largest thorium reserves in the world, primarily found in monazite sands along the coastal regions of Kerala, Tamil Nadu, Odisha, and Andhra Pradesh. According to the Atomic Minerals Directorate for Exploration and Research (AMD), India has nearly 11.93 million tonnes of in situ monazite resources, which contain approximately 1.07 million tonnes of thorium (well over 8 lakh tonnes).
- **Statement 3 is correct:** India's three-stage nuclear power programme, originally envisioned by Dr. Homi Bhabha, is explicitly designed to capitalise on the country's vast thorium reserves. The **third stage** involves Advanced Heavy Water Reactors (AHWRs) that will be powered by a Thorium-232 and Uranium-233 fuel cycle.

- **Statement 4 is correct:** In the second stage of the programme, **Fast Breeder Reactors (FBRs)** are utilised. A “blanket” of Thorium-232 is introduced around the reactor core. The Thorium absorbs excess neutrons generated by the core’s fission process and breeds into Uranium-233. This Uranium-233 is then extracted and reprocessed to serve as the primary fissile fuel for the upcoming third stage.

Q 49. Ans. C

Explanation:

- **Condensation polymers** are formed by repeated condensation reactions between monomer molecules containing two or more functional groups. During polymerisation, **small molecules such as water, hydrogen chloride (HCl), or methanol are eliminated** as by-products. These polymers are also known as **step-growth polymers**.
- **Bakelite** is a condensation polymer formed by the reaction of **phenol** and **formaldehyde**. During its formation, water molecules are eliminated, producing a highly cross-linked, hard, and heat-resistant thermosetting polymer. Because of its excellent electrical insulating properties and mechanical strength, Bakelite is widely used in electrical switches, handles of cooking utensils, and other electrical fittings.
- In contrast, **polypropylene, polyethene, and PVC (polyvinyl chloride)** are **addition polymers**. They are formed by the addition polymerisation of unsaturated monomers such as propene, ethene, and vinyl chloride, respectively, **without the elimination of any small molecule**.

Q 50. Ans. D

Explanation:

- **Ethylene (ethene, C_2H_4)** is one of the most important industrial organic compounds with a wide range of applications.
- **Ethylene is used in the manufacture of mustard gas.** Industrially, ethylene reacts through chlorination and related processes to produce intermediates used in the synthesis of **sulfur mustard (mustard gas)**. This represents a historical and industrial application of ethylene.
- **Ethylene is used to promote fruit ripening.** Ethylene is a **gaseous plant hormone** that accelerates fruit ripening. It is commercially used to induce uniform ripening of fruits such as bananas, mangoes, tomatoes, and apples.
- **Ethylene serves as a raw material for polymer**

production. Ethylene undergoes **addition polymerisation** to form **polyethene (polythene)**, one of the most widely used plastics. It is also an important feedstock for the manufacture of several industrial chemicals.

Q 51. Ans. B

Explanation:

- The **acidity of a base** refers to the **number of ionisable hydroxide (OH^-) ions** that one molecule of the base can furnish (donate) in aqueous solution. In other words, it indicates the maximum number of hydroxide ions that can be released by one formula unit of the base during ionisation.
- **For example:**
 - ✓ **NaOH** releases one OH^- ion and is therefore a **monoacidic base**.
 - ✓ **$Ca(OH)_2$** releases **two OH^- ions** and is a **diacidic base**.
 - ✓ **$Al(OH)_3$** releases **three OH^- ions** and is a **triacidic base**.
- The **number of H^+ ions a base can donate** is incorrect because bases donate OH^- ions, whereas acids donate H^+ ions. Similarly, the **pH value** and **solubility in water** are properties of a base but do not define its acidity.

Q 52. Ans. A

Explanation:

- **Statement 1 is correct:** **Sodium benzoate** is a widely used **food preservative**. It is most effective in **acidic foods and beverages** (generally at pH below 4.5), where it inhibits the growth of bacteria, yeasts, and moulds. It is commonly added to products such as fruit juices, soft drinks, pickles, sauces, jams, and salad dressings to increase their shelf life by preventing microbial spoilage. Therefore, this statement is correct.
- **Statement 2 is incorrect:** Sodium benzoate is **not an artificial sweetener**; it is a **chemical preservative**. Artificial sweeteners such as **saccharin, aspartame, sucralose, and alitame** are used to impart sweetness to foods without adding significant calories. Sodium benzoate has no sweetening function and is used solely to preserve food. Therefore, this statement is incorrect.

Q 53. Ans. A

Explanation:

- **Statement 1 is correct:** **Detergents are more effective than soaps in hard water**. Hard water contains calcium (Ca^{2+}) and magnesium (Mg^{2+})

ions, which react with soaps to form insoluble scum, reducing their cleansing action. Detergents, however, do **not** form insoluble precipitates with these ions and therefore retain their cleansing ability even in hard water. Hence, detergents are preferred for washing with hard water.

- **Statement 2 is incorrect:** Detergents are **synthetic cleansing agents**, most commonly **sodium salts of long-chain alkyl benzene sulphonates or alkyl sulphates**. They contain **sulphonate** ($-\text{SO}_3^-$) or **sulphate** ($-\text{OSO}_3^-$) functional groups rather than ester groups. **Esters of carboxylic acids** are a different class of organic compounds and are not detergents. Therefore, this statement is incorrect.

Q 54. Ans. D

Explanation:

- **Statement 1 is correct:** Brass is an alloy made by combining **copper and zinc**. Alloying improves the properties of metals, such as strength, hardness, corrosion resistance, and workability. Brass is stronger and more resistant to corrosion than pure copper and is commonly used for making screws, musical instruments, decorative items, electrical fittings, and plumbing components. Therefore, this statement is correct.
- **Statement 2 is correct:** **bronze** is an alloy of **copper and tin**. It is harder and more corrosion-resistant than pure copper and has been used since ancient times for making **utensils, statues, coins, medals, bells, and sculptures**. Because of its durability and resistance to wear and corrosion, bronze is suitable for manufacturing utensils. Therefore, this statement is also correct.

Q 55. 55. Ans. A

Explanation:

- **Statement 1 is correct:** **metal carbonates and metal hydrogen carbonates react with acids to form the corresponding salt, carbon dioxide (CO_2), and water**. This reaction is a characteristic property of carbonates and is commonly used to identify them.
- **Statement 2 is correct:** when carbon dioxide is passed through **lime water** [$\text{Ca}(\text{OH})_2$], it first forms **insoluble calcium carbonate** (CaCO_3), turning the solution milky.
 - ✓ $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
- On passing **excess carbon dioxide**, the insoluble calcium carbonate dissolves to form **soluble calcium bicarbonate** [$\text{Ca}(\text{HCO}_3)_2$], causing the milkiness to disappear.
 - ✓ $\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{HCO}_3)_2$

- **Statement 3 is incorrect:** Carbon dioxide **does react with lime water**. Initially, it turns lime water **milky** due to the formation of calcium carbonate. On passing excess carbon dioxide, the milkiness disappears because soluble calcium bicarbonate is formed. Hence, carbon dioxide clearly affects lime water, making this statement incorrect.

Q 56. Ans. C

Explanation:

- **Statement 1 is correct:** the **octane number** is a measure of the **anti-knocking quality** of petrol (gasoline). It indicates the fuel's resistance to knocking or premature ignition in a spark-ignition (petrol) engine. A **higher octane number** means greater resistance to knocking and better engine performance. Therefore, this statement is correct.
- **Statement 2 is incorrect:** the **cetane number** is used to measure the **ignition quality of diesel fuel**, i.e., how readily diesel ignites under compression in a compression-ignition engine. It does **not** measure the **volatility** of a fuel. Volatility refers to the tendency of a liquid to vaporise, which is a different physical property. Therefore, this statement is incorrect.
- **Statement 3 is correct:** **iso-octane** (2,2,4-trimethylpentane) is assigned an **octane number of 100** because of its excellent resistance to knocking. In contrast, **n-heptane**, which knocks very easily, is assigned an octane number of **0**. The octane number of a petrol sample is determined by comparing its knocking characteristics with mixtures of iso-octane and n-heptane. Therefore, this statement is correct.

Q 57. Ans. B

Explanation:

- **Glycan Atlasing** is a breakthrough imaging technique developed by researchers at the **Max Planck Institute for the Science of Light**. It combines **DNA-tagged lectin labelling** with **super-resolution microscopy** to map the **glycocalyx** (sugar coating on cell surfaces) at nanometer-scale resolution. By detecting changes in **glycan (sugar) patterns**, the technique can identify **early-stage cancer**, distinguish **healthy and diseased cells**, and monitor **immune responses**, offering a powerful tool for disease diagnosis and biomedical research.

Q 58. Ans. A

Explanation:

- **Statement:** Glycolysis is the first stage of cellular

respiration and takes place in the **cytoplasm (cytosol)** of the cell. During this process, one molecule of glucose is broken down into two molecules of pyruvate through a series of enzyme-catalysed reactions. Glycolysis is a universal pathway that occurs in the cytoplasm of both prokaryotic and eukaryotic cells, irrespective of whether oxygen is present.

- **Conclusion I:** Glycolysis is a **common pathway for both aerobic and anaerobic respiration**. It occurs in **all living organisms**, including aerobic organisms that use oxygen and anaerobic organisms that survive without oxygen. After glycolysis, the fate of pyruvate depends on the availability of oxygen, but glycolysis itself is **not restricted to aerobic organisms**. Therefore, **Conclusion I is incorrect**.
- **Conclusion II:** glycolysis takes place entirely in the **cytoplasm**, not in the mitochondria. The mitochondrial matrix is the site of the **Krebs cycle (citric acid cycle)**, while glycolysis is completed before pyruvate enters the mitochondria in aerobic cells. Hence, the statement that glycolysis occurs only inside mitochondria is incorrect, making **Conclusion II incorrect**.

Q 59. Ans. D

Explanation:

- **Nafithromycin** is **India's first indigenously developed antibiotic**, launched to address the growing challenge of **antimicrobial resistance (AMR)**. Developed by **Wockhardt** in collaboration with the government's **Biotechnology Industry Research Assistance Council (BIRAC)**, it is a **novel macrolide/ketolide antibiotic** designed to treat **drug-resistant respiratory infections**. Marketed under the brand name **Miqnaf**, the drug is particularly effective against **Community-Acquired Bacterial Pneumonia (CABP)** and offers the advantage of a **short 3-day treatment regimen**.

Q 60. Ans. B

Explanation:

- **Aditya Pandya**, aged 17, became **India's youngest analogue astronaut** after successfully completing an **eight-day simulated lunar habitat mission** at **Dholavira, Gujarat**. Organised by **AAKA Space Studio** from **1–8 February 2026**, the mission placed a four-member crew inside a container-based lunar habitat to simulate space-living conditions. In addition to serving as a crew member, Aditya developed the habitat's **Internet of Things (IoT)**

systems, air-quality monitoring sensors, and biometric monitoring tools, contributing to the mission's engineering and research objectives.

Q 61. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The **Global Hunger Index (GHI)** is a peer-reviewed annual report jointly published by the European NGOs **Concern Worldwide (Ireland)** and **Welthungerhilfe (Germany)**. It tracks hunger at global, regional, and national levels.
- **Pair 2 is correctly matched:** The **Human Development Report (HDR)** is an annual flagship report published by the Human Development Report Office of the **United Nations Development Programme (UNDP)**. It introduces the **Human Development Index (HDI)**.
- **Pair 3 is incorrectly matched:** The **World Competitiveness Ranking (or Index)** is published annually by the **Institute for Management Development (IMD)** based in Switzerland, not the International Management Institute (IMI).
- **Pair 4 is incorrectly matched:** The **Global Innovation Index (GII)** is co-published by the **World Intellectual Property Organization (WIPO)** in partnership with the **Portulans Institute** (and historically **INSEAD** and **Cornell University**). The **World Economic Forum (WEF)**, on the other hand, publishes reports like the **Global Competitiveness Report** and the **Global Gender Gap Report**.

Q 62. Ans. C

Explanation:

- **Bharatmala Pariyojana** is an umbrella highway development program managed by the Ministry of Road Transport and Highways (**MoRTH**). Its primary focus is optimizing the efficiency of freight and passenger movement across the country by bridging critical infrastructure gaps.
- **Rural Road Connectivity under PMGSY** (**Pradhan Mantri Gram Sadak Yojana**) is **not** a component of **Bharatmala**. **PMGSY** is an entirely separate, older scheme launched in 2000 under the Ministry of Rural Development, specifically aimed at providing all-weather road connectivity to unconnected rural habitations.
- **The official components of Bharatmala Pariyojana Phase I include:**
 - ✓ **Economic Corridors** (Development of ~9,000 km)

- ✓ **Inter-corridor and Feeder Roads** (Development of ~6,000 km)
- ✓ **National Corridors Efficiency Improvement** (Development of ~5,000 km)
- ✓ **Border and International Connectivity Roads** (Development of ~2,000 km)
- ✓ **Coastal and Port Connectivity Roads** (Development of ~2,000 km)
- ✓ **Expressways** (Development of ~800 km)

Q 63. Ans. A

Explanation:

- **Statement 1 is correct:** In 2025, the Union Cabinet approved the **Export Promotion Mission (EPM)** with a total outlay of Rs 25,060 crore. It is a comprehensive six-year initiative spanning FY 2025-26 to FY 2030-31, designed to consolidate multiple fragmented schemes and boost India's export competitiveness, especially for MSMEs, first-time exporters, and labor-intensive sectors.
- **Statement 2 is correct:** India's engineering goods sector witnessed a landmark year in FY 2024-25, hitting an all-time high of **US\$ 116.7 billion** in exports. With total merchandise exports standing at roughly US\$ 437.4 billion for that fiscal year, engineering goods emerged as the largest contributor, accounting for **26.67%** of the total merchandise export basket.
- **Statement 3 is correct:** India's top five merchandise export destinations are the **United States, United Arab Emirates (UAE), Netherlands, United Kingdom, and China**. Both the UAE and China maintained their positions within the top five during the 2025-26 period, with China emerging as one of the fastest-growing major export destinations for India.
- **Statement 4 is incorrect:** While India is a formidable player in global trade, it did not rank in the top 10 for merchandise exports. According to the World Trade Organization (WTO) and the Directorate General of Commercial Intelligence and Statistics (DGCI&S), India ranked **18th globally in merchandise exports** in 2024. However, India does rank in the top 10 (8th position) for commercial services exports.

Q 64. Ans. A

Explanation:

- **Statement 1 is correct:** A derivative is a contract between two or more parties whose value is based on an underlying asset such as stocks, bonds,

commodities, currencies, interest rates, or market indices.

- **Statement 2 is incorrect:** The value of a derivative depends on the price or value of the underlying asset; it is not independent of it.

Q 65. Ans. A

Explanation:

- **The Balance of Payments (BoP) is broadly divided into two main accounts:** the **Current Account** and the **Capital Account**.
- **Exports:** The export (and import) of goods and services is the primary component of the Current Account. Trade in physical goods forms the "Balance of Trade," while trade in services forms part of the "Invisibles."
- **Private Transfers:** These include unilateral transfers such as remittances sent by Indians working abroad, gifts, and grants. Because these do not create any future repayment obligation or liability, they are recorded under the "Invisibles" section of the Current Account.
- **Net Commercial Borrowings:** External Commercial Borrowings (ECBs) and other loans involve borrowing money from foreign entities. Because they create a future liability (debt that must be repaid), they are recorded in the **Capital Account**.
- **Foreign Direct Investment:** FDI (along with Foreign Portfolio Investment or FPI) represents a flow of capital used to purchase assets or stakes in Indian companies. Since it deals with investments and changes in asset ownership, it is recorded in the **Capital Account**.

Q 66. Ans. A

Explanation:

- **Statement 1 is correct:** Tax buoyancy is a key macroeconomic indicator that measures how responsive a government's tax revenue is to changes in the country's Gross Domestic Product (GDP). It is calculated as the percentage change in tax revenue divided by the percentage change in GDP. For example, a tax buoyancy greater than 1 indicates that tax revenues are growing at a faster rate than the overall economy.
- **Statement 2 is incorrect:** This statement directly contradicts the core concept of tax buoyancy. Tax buoyancy explicitly demonstrates that tax revenue is highly related to and dependent on the rate of economic growth. The entire purpose of the metric is to quantify exactly how connected they are.

Q 67. Ans. B

Explanation:

- A surcharge is fundamentally an additional charge, tax, or fee added to the standard or base price of a good, service, or existing tax. It is explicitly an **extra cost** that is applied on top of what is initially quoted or calculated.
- **Common examples of surcharges include:**
 - ✓ A “fuel surcharge” added by airlines or shipping companies to cover sudden spikes in oil prices.
 - ✓ A “late fee surcharge” on a utility bill.
 - ✓ An income tax surcharge (a tax levied on the tax payable, not on the total income directly) for individuals earning above a certain high-income threshold.

Q 68. Ans. C

Explanation:

- According to the Economic Survey 2025–26, India’s services exports touched an all-time high of **USD 387.6 billion in FY25**, registering a robust year-on-year growth of **13.6%**.
- This strong performance reinforces India’s position as a global hub for technology, business, and professional services, helping the country become the world’s seventh-largest services exporter. Total overall exports (merchandise + services) for FY25 reached a record **USD 825.3 billion**.

Q 69. Ans. B

Explanation:

- According to a Discussion Paper released by NITI Aayog in January 2024 titled “Multidimensional Poverty in India since 2005–06,” India has registered a steep and significant decline in multidimensional poverty.
- The multidimensional poverty headcount ratio dropped dramatically from **55.34%** in 2005–06 to an estimated **11.28%** in 2022–23.
- This steep decline means that an estimated **24.82 crore (248.2 million) individuals** successfully escaped multidimensional poverty over the nine-year period between 2013–14 and 2022–23 alone.
- **The NITI Aayog National MPI assesses poverty across three equally weighted dimensions:** Health, Education, and Standard of Living, using 12 sustainable development goals-aligned indicators.

Q 70. Ans. A

Explanation:

- **The Sachetisation Plan** (introduced in a consultation

paper by SEBI in 2025) explicitly aims to promote financial inclusion by breaking down mutual fund investments into highly affordable “sachets”, similar to how consumer goods like shampoo or coffee are sold in small packets to increase rural and low-income market penetration.

- Under this initiative, SEBI proposed **small-ticket Systematic Investment Plans (SIPs) starting at just ₹250 per month**.
- The primary target demographic is **low-income groups, daily wage earners, rural households, and first-time investors** who previously found traditional mutual fund minimums out of reach. By lowering the entry barrier, SEBI aims to inculcate a habit of systematic saving and democratize wealth creation for the underserved sections of the Indian economy.

Q 71. Ans. C

Explanation:

- The Charter Act of 1833 ended the Company’s remaining trade monopoly, including its monopoly of trade with China and of the tea trade, which the Charter Act of 1813 had left untouched. The Company thereby ceased to be a commercial body and became a purely administrative one, holding its territories in trust for the Crown.
- The 1833 Act also made the Governor General of Bengal the Governor General of India, the first incumbent being Lord William Bentinck, and it centralised legislative power in the Governor General in Council while the Governors of Bombay and Madras lost their legislative powers. It attempted an open competition principle for civil service recruitment, but this was negated by the Court of Directors, leaving the Charter Act of 1853 to actually introduce open competition.

Q 72. Ans. A

Explanation:

- **a is 2:** The Regulating Act of 1773 provided for the establishment of a Supreme Court at Calcutta, comprising a chief justice and three other judges, and it also designated the Governor of Bengal as the Governor General of Bengal, subordinating the Governors of Bombay and Madras to him.
- **b is 1:** The Charter Act of 1853 separated, for the first time, the legislative and executive functions of the Governor General’s council by adding six new legislative councillors, effectively creating a distinct Central Legislative Council.
- **c is 3:** The Government of India Act of 1919,

the Montagu-Chelmsford Reforms, introduced dyarchy in the provinces by dividing subjects into transferred subjects administered with responsible ministers and reserved subjects administered without such responsibility.

- **d is 4:** The Government of India Act of 1935 abolished this provincial dyarchy and introduced provincial autonomy, while attempting to introduce dyarchy at the Centre, a scheme that never came into operation because the all-India federation for which the Act provided did not materialise.

Q 73. Ans. B

Explanation:

- **Assertion is true:** The Objectives Resolution, moved by Jawaharlal Nehru on 13 December 1946, laid down the fundamentals and underlying philosophy of the constitutional structure, and its ideals were later incorporated substantially into the Preamble of the Constitution.
- **Reason is true:** The Objectives Resolution was debated at length in the Constituent Assembly and was formally adopted on 22 January 1947.
- However, the Reason does not explain the Assertion. The fact that the Resolution was adopted on 22 January 1947 is merely a historical date and has no causal connection to why it later formed the basis of the Preamble. The Preamble was shaped by the content and ideals of the Resolution, not by the date of its adoption. Since both statements are independently true, but one does not logically explain the other, the correct answer is B.

Q 74. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Article 40, directing the organisation of village panchayats, is classified as a Gandhian principle, reflecting Gandhi's vision of village self-government.
- **Pair 2 is correctly matched:** Article 44, on securing a uniform civil code, is classified as a liberal intellectual principle, reflecting a modernising and rationalist strand within the directive principles.
- **Pair 3 is correctly matched:** Article 39A, directing the promotion of equal justice and free legal aid, is a socialist principle, and it was one of the directives added by the 42nd Amendment Act of 1976.
- **Pair 4 is incorrectly matched:** Article 48A, directing the protection and improvement of the environment and the safeguarding of forests and

wildlife, is not classified as a Gandhian principle; it is a directive added by the 42nd Amendment that classifies under the liberal intellectual principles.

Q 75. Ans. B

Explanation:

- **Pair 1 is correct:** Article 23 prohibits traffic in human beings and forced labour, popularly termed begar, and permits the State to impose compulsory service for public purposes without discrimination on grounds only of religion, race, caste, or class.
- **Pair 2 is correct:** Article 24 prohibits the employment of children below the age of fourteen years in any factory, mine, or other hazardous employment, a protection that operates independently of any implementing legislation.
- **Pair 3 is correct:** Article 20 protects against conviction under an ex post facto criminal law, that is, a law that retrospectively declares an act to be an offence, along with protection against double jeopardy and self-incrimination.
- **Pair 4 is incorrect:** Article 22 deals with protection against arrest and detention, including rights on arrest and safeguards regarding preventive detention, not with double jeopardy; the protection against double jeopardy is actually found in Article 20(2), so this pair mismatches the

Q 76. Ans. C

Explanation:

- In the I. R. Coelho case of 2007, the Supreme Court held that laws placed in the Ninth Schedule after 24 April 1973, the date of the Kesavananda Bharati judgment, are open to judicial review if they violate the fundamental rights that form part of the basic structure of the Constitution, thereby removing the absolute immunity that Article 31B had earlier conferred on that schedule.
- The Ninth Schedule was originally added by the 1st Amendment Act of 1951 to protect land reform laws from challenge under Article 13. The Golak Nath case of 1967 had earlier held that fundamental rights cannot be amended at all, a position the Kesavananda Bharati case of 1973 reversed by permitting amendment of fundamental rights subject to the basic structure, while the Minerva Mills case of 1980 added the harmony between fundamental rights and directive principles, and the power of judicial review itself, to that basic structure.

Q 77. Ans. A**Explanation:**

- **a is 2:** The absolute veto, under which the President withholds assent and the bill lapses altogether, is normally exercised over private member bills, and also over government bills when the council of ministers resigns before the President acts and the new ministry advises withholding assent.
- **b is 3:** The suspensive veto allows the President to return a bill, other than a money bill, for reconsideration; if Parliament passes it again with or without amendments, the President must give assent, so this veto is overridden by an ordinary majority on reconsideration.
- **c is 1:** The pocket veto, where the President neither ratifies, rejects, nor returns a bill, was exercised by President Zail Singh in 1986 on the Indian Post Office Amendment Bill, since the Constitution prescribes no time limit for the President to act.
- **d is 4:** The qualified veto, which can be overridden by the legislature with a higher special majority, is a power possessed by the American President and is not possessed by the Indian President, who has no such intermediate category of veto.

Q 78. Ans. A**Explanation:**

- **Assertion is true:** The National Judicial Appointments Commission, intended to replace the collegium system for appointing judges to the higher judiciary, never became operative for making appointments, since it was struck down before it could function in that capacity.
- **Reason is true and correctly explains the Assertion:** In 2015 the Supreme Court struck down the 99th Constitutional Amendment Act of 2014, which had created the Commission, as unconstitutional on the ground that it violated the independence of the judiciary forming part of the basic structure, and this striking down is precisely why the Commission never functioned, so the Reason correctly explains the Assertion.
- The collegium system, which has no express mention in the Constitution, was consequently revived and continues to operate through the three judges cases of 1981, 1993, and 1998, under which the Chief Justice of India and a body of senior judges control appointments and transfers of judges of the Supreme Court and the High Courts, with the President formally appointing judges under Article 124 on the collegium's recommendation.

Q 79. Ans. C**Explanation:**

- **Pair 1 is correctly matched:** The Cabinet Committee on Security is a standing committee, one of the permanent bodies dealing with matters of national security and defence, and it is chaired by the Prime Minister.
- **Pair 2 is incorrectly matched:** The Appointments Committee of the Cabinet is a standing committee, not an ad hoc one, since it permanently handles appointments to senior posts in the government and public sector undertakings, unlike ad hoc committees, which are temporary and set up for specific purposes.
- **Pair 3 is correctly matched:** The Cabinet Committee on Economic Affairs is a standing committee that reviews economic trends and issues affecting economic policy, and along with the Cabinet Committee on Security, is one of the more prominent standing committees.
- **Pair 4 is correctly matched:** Groups of Ministers, like cabinet committees generally, are extra constitutional in emergence, being set up under the Rules of Business rather than under any express constitutional provision, so three of the four pairs are correctly matched.

Q 80. Ans. B**Explanation:**

- **Pair 1 is correct:** In the Golak Nath case of 1967, the Supreme Court held that Parliament cannot amend the fundamental rights, treating a constitutional amendment as law within the meaning of Article 13, a position later reversed in the Kesavananda Bharati case.
- **Pair 2 is correct:** In the Minerva Mills case of 1980, the Court added the power of judicial review and the balance between fundamental rights and directive principles to the basic structure, while striking down provisions of the 42nd Amendment that had sought to place amendments beyond judicial scrutiny.
- **Pair 3 is correct:** In the Maneka Gandhi case of 1978, the Court held that the procedure established by law under Article 21 must be fair, just, and reasonable, effectively reading the substance of due process into the article and linking it with Articles 14 and 19.
- **Pair 4 is incorrect:** The Berubari Union case of 1960 actually held that the Preamble is not a part of the Constitution; it was the later Kesavananda Bharati case of 1973 that reversed this position and held the Preamble to be a part of the Constitution, so this pair wrongly attributes the later holding to the Berubari case.

Q 81. Ans. C**Explanation:**

- According to the **2011 Census**, Patna is the most populous district of Bihar, with a population exceeding **5.8 million**. It also records the **highest population density** among the top five districts at **1,823 persons per sq. km**, and leads in education with a **literacy rate of 70.68%**.
- **East Champaran** ranks second in terms of population, crossing the **5 million mark**. However, it has a comparatively lower population density of **1,285 persons per sq. km** and the **lowest literacy rate (55.79%)** among the top five districts.
- **Muzaffarpur** stands third, with a population of around **4.8 million**. It has a relatively high population density of **1,514 persons per sq. km** and a **literacy rate of 63.43%**.

Q 82. Ans. C**Explanation:**

- The **State Goods and Services Tax (SGST)** is projected to be the **largest contributor to Bihar's own tax revenue** in the **2026–27 financial year**. As per the **Bihar Budget Analysis 2026–27**, SGST is expected to account for nearly **58% of the state's total internal tax collection**, highlighting the state's strong dependence on a **consumption-driven revenue structure**.
- Additionally, SGST collections in Bihar are anticipated to show **robust growth**, with an estimated **annual increase of around 14%** compared to the actual revenue figures of **2024–25**.

Q 83. Ans. D**Explanation:**

- **Rs 21,270 crore** is the exact allocation designated for the **health sector** in the **Bihar Budget 2026-27**, proposed by Finance Minister **Bijendra Prasad Yadav**. This funding focuses primarily on building infrastructure, converting district clinics into advanced medical hubs, and constructing **10 new medical colleges**.

Q 84. Ans. A**Explanation:**

- **Assertion (A) is true:** Bihar has been divided into **three agro-climatic zones** to facilitate **region-specific agricultural planning and resource management**. The division recognizes that agricultural conditions differ significantly across the state, requiring distinct cropping patterns, irrigation strategies, and farming practices.

- **Reason (R) is true:** agro-climatic zones of Bihar differ in terms of **soil characteristics, rainfall distribution, temperature, topography (terrain), groundwater availability, and flood-drought conditions**. These variations directly influence crop suitability, agricultural productivity, and water management. Therefore, separate planning units have been created to ensure effective agricultural development.
- Since the **agro-climatic zoning of Bihar is based precisely on these variations in natural and climatic factors**, the Reason correctly explains the Assertion.

Q 85. Ans. A**Explanation:**

- **Assertion (A) is true:** Bihar's agriculture benefits from **highly fertile alluvial soils**, abundant river systems, and a favourable monsoon climate, making the state one of India's leading producers of crops such as **rice, wheat, maize, pulses, and horticultural produce**. At the same time, agriculture faces constraints such as **annual floods in north Bihar, drought conditions in parts of south Bihar, soil nutrient depletion due to intensive cultivation, waterlogging, and climatic variability**.
- **Reason (R) is true:** The **alluvial soils** of Bihar are naturally fertile and support intensive agriculture. However, **continuous cropping, inadequate replenishment of nutrients, and imbalanced fertilizer use** have led to **declining soil fertility in many areas**. In addition, Bihar experiences **season-specific weather stresses**, including **floods during the southwest monsoon, drought in parts of south Bihar, heat waves during summer, and cold waves in winter**, all of which affect agricultural productivity.
- The Reason directly explains the Assertion, because the same soil and climatic conditions that enhance agricultural potential also create significant production constraints.

Q 86. Ans. D**Explanation:**

- **Maulana Mazharul Haque (1866–1930)** was born on **22 December 1866** in **Brahmpur village, Maner** (present-day Patna district), Bihar. In **1900**, he settled in **Faridpur village** of present-day **Siwan district** on land donated by his relatives. A prominent nationalist leader, he was elected **Vice-Chairman of the Bihar Congress Committee** in

1906 and also started the journal **Motherland**. He actively participated in the **Champaran Satyagraha**, for which he was sentenced to **three months' imprisonment**. Maulana Mazharul Haque founded the **Sadaquat Ashram** in Patna and the **Bihar Vidyapeeth**, both of which played an important role in India's freedom movement and national education.

Q 87. Ans. B

Explanation:

- **Baikunth Shukla** was a fearless revolutionary from **Vaishali, Bihar**, remembered for assassinating **Phanindra Nath Ghosh**, the approver whose testimony led to the execution of **Bhagat Singh, Rajguru, and Sukhdev**. He carried out the assassination on **9 November 1932** as an act of retribution against British rule. Following his arrest, he was tried by the colonial government and sentenced to death. **Baikunth Shukla was executed by hanging at Gaya Central Jail on 14 May 1934**, at the age of 27, just one day before his 28th birthday.

Q 88. Ans. C

Explanation:

- **Statement 1 is correct:** **Babu Veer Kunwar Singh**, the **zamindar of Jagdishpur** (present-day **Bhojpur district**), emerged as the foremost leader of the **Revolt of 1857** in Bihar. After the **Danapur uprising on 25 July 1857**, the rebel sepoys joined him at Jagdishpur, from where he led the resistance against the British.
- **Statement 2 is incorrect:** **Veer Kunwar Singh** was born in 1777. When the Revolt of 1857 began, he was **around 80 years old**, not 59. Despite his advanced age, he led the rebellion with remarkable courage and military skill.
- **Statement 3 is correct:** The Government of Bihar observes **Veer Kunwar Singh Vijayotsav** every year on **23 April** to commemorate **Kunwar Singh's victory at the Battle of Jagdishpur on 23 April 1858**, when he defeated the British forces led by Captain **Le Grand** and recaptured Jagdishpur.

Q 89. Ans. A

Explanation:

- **Statement 1 is correct:** **Pir Ali Khan**, one of the earliest martyrs of the **Revolt of 1857** in Bihar, was **executed by the British on 7 July 1857** in Patna for his active role in organizing and leading the uprising against British rule.

- **Statement 2 is correct:** **Pir Ali Khan** was a **bookseller and revolutionary based in Patna**. He mobilized local revolutionaries and played a key role in the uprising in the city. His activities made Patna an important centre of resistance during the Revolt of 1857.
- **Statement 3 is incorrect:** The Government of Bihar observes **Shaheed Pir Ali Diwas on 7 July**, the anniversary of his martyrdom, **not on 7 June**.

Q 90. Ans. A

Explanation:

- **Statement 1 is correct:** The **Kosi River** enters Bihar from **Nepal** near **Bhimnagar** in **Supaul district**. After entering the state, it flows through the Kosi plains, influencing the districts of Supaul, Saharsa, Madhepura, Khagaria and adjoining areas.
- **Statement 2 is correct:** The **Kosi Barrage**, also known as the **Bhimnagar Barrage**, is located at **Bhimnagar** on the Indo–Nepal border. Constructed under the **Kosi Project**, it regulates river flow, provides irrigation, and helps in flood management.
- **Statement 3 is incorrect:** The **Kosi River** does not join the **Ganga** in **Bhagalpur district**. The river finally meets the **Ganga** near **Kursela** in **Katihar district**. The Kosi is often called the “**Sorrow of Bihar**” because of its frequent floods and historical tendency to **shift its course westward over a vast alluvial fan**. Over the past two centuries, the river has migrated by more than **100 km**, causing widespread changes in drainage patterns and severe flood damage across north Bihar.

Q 91. Ans. B

Explanation:

- The **Burhi Gandak River** originates from the **Someshwar Hills** (also called the **Someshwar Range**) in the foothills of the Himalayas in **West Champaran district** near the Indo–Nepal border. From its source, it flows southeast through the districts of **West Champaran, East Champaran, Muzaffarpur, Samastipur, Begusarai, and Khagaria**, before joining the **Ganga** near **Khagaria**. It is one of the major north Bihar rivers and forms a separate river basin under the **Second Bihar State Irrigation Commission (1994)**.

Q 92. Ans. C

Explanation:

- **Nagi Dam Wildlife Sanctuary**, located in **Jamui district**, is a **bird sanctuary** centered around the

Nagi Dam reservoir. It is renowned for hosting large numbers of migratory water birds during the winter season. Species such as Bar-headed Goose, Northern Pintail, Common Teal, Gadwall, Eurasian Wigeon, and Red-crested Pochard visit the sanctuary, making it one of Bihar's important wetland bird habitats. Owing to its ecological importance, Nagi-Nakti Bird Sanctuary has also been designated as a Ramsar Site.

Q 93. Ans. C

Explanation:

- Founded in 1632, the Dutch Factory at Patna served as a major trading station of the Dutch East India Company (VOC), dealing primarily in saltpeter, opium, and textiles. The surviving 17th-century building, located within the Patna College campus on Ashok Rajpath, is one of the few remaining examples of Dutch colonial architecture in Bihar.
- The factory and its adjoining warehouses operated as an important inland commercial hub until they were transferred to the British under the Anglo-Dutch Treaty (Treaty of London) of 1824. Today, the structure remains an important symbol of Patna's colonial trading legacy and Bihar's historic river-port economy.

Q 94. Ans. B

Explanation:

- Statement 1 is incorrect: the Kosi-Mechi River Link Project proposes to transfer surplus water from the Kosi River to the Mechi River, not from the Mechi to the Kosi. The project is part of the National Perspective Plan for Interlinking of Rivers (1980), which seeks to facilitate inter-basin transfer of water to improve irrigation and regional water management.
- Statement 2 is correct: The project is expected to generate significant irrigation benefits in northeastern Bihar. The proposed command area includes the districts of Araria, Purnea, Kishanganj, and Katihar. Overall, the project aims to provide annual irrigation to about 4.74 lakh hectares, of which nearly 2.99 lakh hectares lie in Bihar.
- Statement 3 is correct: The project is specifically designed to support irrigation in the Mahananda basin, where it is expected to provide Kharif-season irrigation to about 2.15 lakh hectares. Besides irrigation, it also proposes to supply around 24 million cubic metres of water annually for domestic and industrial use, thereby improving overall water availability in the region.

Q 95. Ans. B

Explanation:

- The Patna High Court was established by Letters Patent and officially began functioning on 3 February 1916 with one Chief Justice and six puisne judges. Its first Chief Justice was Sir Edward Maynard Des Champs Chamier. Initially, the court exercised civil, criminal, admiralty, and matrimonial jurisdiction over the then provinces of Bihar and Orissa. Over the years, its jurisdiction and sanctioned strength of judges have expanded considerably. Justice Meenakshi Madan Rai assumed office as the Chief Justice of the Patna High Court on 5 June 2026, following her appointment by the President of India, Droupadi Murmu.

Q 96. Ans. D

Explanation:

- Bihar is the largest mushroom-producing state in India, contributing over 11% of the country's total production. It is followed by Maharashtra and Odisha as the next leading mushroom-producing states. The sector's growth has been supported by favourable climatic conditions, abundant availability of substrates such as paddy straw, and dedicated state-level agricultural initiatives. While Bihar leads in overall production, Maharashtra and Odisha are also important centres for the commercial cultivation of Button, Oyster, and Milky mushrooms.

Q 97. Ans. A

Explanation:

- Sumitra Rani Saha (60), a resident of Ara in Bhojpur district, became the first person in Bihar to receive Indian citizenship under the Citizenship (Amendment) Act (CAA), 2019. She was granted her citizenship certificate on 3 January 2025, bringing to an end a 40-year wait after migrating to India. The certificate was presented following a meeting of the State-Level Empowered Committee at the Bihar Secretariat, Patna, chaired by M. Ramachandran, Director of Census and Citizen Registration.

Q 98. Ans. C

Explanation:

- The 18th Bihar Legislative Assembly, constituted after the 2025 Assembly elections, includes 29 women MLAs. This represents an increase from

the 26 women legislators (10.7%) elected to the previous Assembly in 2020, indicating a gradual rise in women's representation in the state legislature.

- Bihar continues to witness **higher voter turnout among women than men**. In the 2025 Assembly elections, **female voter turnout was nearly 60%**, compared with **about 54% for male voters**, reflecting the growing political participation of women in the state.
- The newly elected women legislators are relatively young in age. Among the **29 women MLAs**, **13 are between 25 and 39 years**, while **9 belong to the 40–54 years age group**, highlighting the increasing presence of younger women in Bihar's legislative politics.
- The educational profile of the elected women MLAs reveals considerable diversity. **Around half of them (15 out of 29) do not possess a college degree**, indicating that electoral success in Bihar is influenced not only by formal educational qualifications but also by factors such as political experience, grassroots engagement, social influence, and public support.

Q 99. Ans. D

Explanation:

- Bihar is set to establish its **first two Special Economic Zones (SEZs)** with an investment of **₹225 crore** to promote **export-led industrial growth** and attract investment. The proposed SEZs will be developed at **Nawanagar (Buxar district)** and **Kumarbagh (West Champaran district)**, focusing on **export-oriented industries** such as **textiles, information technology (IT), and electronics**. The proposal received approval from the **Central Government by November 2024**.
- **Special Economic Zones (SEZs)** are designated areas that offer **fiscal incentives, duty-free benefits, and a liberal regulatory framework** to encourage exports, investment, industrial development, and employment generation. In India, **Asia's first Export Processing Zone (EPZ)** was established at **Kandla, Gujarat, in 1965**. The **Special Economic Zones Act, 2005** came into force along with the **SEZ Rules, 2006**, providing the legal framework for SEZ development. At present, **379 SEZs have been notified across India**, of which **265 are operational**. Nearly **64%** of these operational SEZs are concentrated in **Tamil Nadu, Telangana, Karnataka, Andhra Pradesh, and Maharashtra**.

Q 100. Ans. D

Explanation:

- **Statement 1 is correct:** The **Aryabhata Drishti** is a major **technology and innovation project approved by the Bihar State Cabinet**. The project has been sanctioned with a **budget of ₹209 crore** and is being implemented in partnership with the **Singapore-based Global Finance & Technology Network (GFTN)** to strengthen Bihar's innovation and startup ecosystem.
- **Statement 2 is correct:** The project proposes to establish the **Aryabhata Technology Observatory**, an **AI-driven digital sandbox** designed to support **more than 100 startups**. In addition, it aims to train **around 7,000 STEM and non-STEM graduates** through a **five-month Advanced AI Certification Programme** in collaboration with the **National University of Singapore (NUS-AIDF)** and the **Linux Foundation**.

Q 101. Ans. C

Explanation:

- The **Taiga (Boreal Forest) biome**, found across high-latitude regions of **Russia, Canada, and Scandinavia**, experiences long, severely cold winters, often with temperatures dropping below **-30°C to -50°C**, lasting six months or more. Owing to the harsh climate, the Taiga has a short growing season, typically only **3 to 4 months (summer)**, which restricts the biome largely to cold-hardy coniferous trees (spruce, fir, pine, larch) rather than broad-leaved species.
- Continuous, permanently frozen ground (permafrost) throughout the region is the defining feature of the **Tundra biome**, not the Taiga. While some northern/high-latitude fringes of the Taiga do experience discontinuous or patchy permafrost, it is not a universal or defining feature of the biome as a whole, unlike the Tundra, where permafrost underlies almost the entire region and severely restricts drainage and root penetration.

Q 102. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The **Asan Conservation Reserve**, Uttarakhand's first Ramsar site, is located at the confluence of the **Asan and Yamuna rivers** in **Dehradun district**, and officially falls within the **Yamuna River Basin**.
- **Pair 2 is correctly matched:** The **Keshopur-Miani**

Community Reserve lies in the former floodplains of the Beas and Ravi rivers in Gurdaspur, Punjab, both tributaries of the Indus River system.

- **Pair 3 is incorrectly matched:** Nandur Madhameshwar, Maharashtra's first Ramsar site, was formed by a weir at the confluence of the Godavari and Kadwa rivers in Nashik district, it belongs to the Godavari basin, not the Krishna basin.
- **Pair 4 is incorrectly matched:** Coringa Wildlife Sanctuary, near Kakinada, Andhra Pradesh, lies on the deltaic branches of the Godavari and Gautami rivers and is explicitly part of the Godavari estuary, not the Kaveri basin.

Q 103. Ans. C

Explanation:

- During the last Ice Age (Pleistocene epoch), sea levels dropped significantly as vast quantities of water were locked up in continental ice sheets. This exposed the shallow continental shelf beneath what is now the Bering Strait, creating a wide land bridge commonly known as Beringia connecting Siberia (Asia) to Alaska (North America). This land bridge is widely accepted by anthropologists and geologists as the route by which early humans first migrated from Asia into the Americas, roughly 20,000–13,000 years ago.

Q 104. Ans. D

Explanation:

- **Hugli Region → 2 (Jute industry):** The Hugli industrial region, stretching along the banks of the Hugli River in West Bengal, developed historically around the jute industry, owing to its proximity to jute-growing areas of the Ganga delta, cheap water transport, and colonial-era port infrastructure at Kolkata.
- **Mumbai–Pune Region → 4 (Cotton textiles and entertainment industry):** This region grew initially as India's premier cotton textile hub, aided by a humid climate favourable for spinning and easy access to raw cotton from the Deccan. It also hosts India's entertainment/film industry (Bollywood), along with diversified engineering, chemical, and IT industries.
- **Chota Nagpur Region → 1 (Heavy metallurgical industries):** This mineral-rich plateau (Jharkhand, parts of Odisha, West Bengal, Chhattisgarh) has abundant iron ore, coal, and manganese, making it the base for India's heavy metallurgical and iron-and-steel industries — including Jamshedpur (Tata Steel), Bokaro, Durgapur, and Rourkela.

- **Gujarat Region → 3 (Petrochemical complexes):** Gujarat hosts major petrochemical complexes at Jamnagar (Reliance's refinery, one of the world's largest), Vadodara, and Ankleshwar, supported by offshore and onshore crude oil/natural gas reserves along its coast.

Q 105. Ans. A

Explanation:

- **Snow leopard, Ibex, and Tibetan antelope (Chiru)** are all high-altitude cold desert species naturally adapted to the harsh, mountainous, Trans-Himalayan ecosystem of Ladakh. Other prominent wildlife in this region includes the Yak, Tibetan wild ass (Kiang), and Blue sheep (Bharal).
- **Option B is incorrect** because the one-horned rhinoceros is native to the floodplains of the Eastern Himalayas (like Kaziranga in Assam), and the red panda is found in the temperate broadleaf and mixed forests of the Eastern Himalayas (Sikkim, Arunachal Pradesh, Darjeeling).
- **Option C is incorrect** because the Asiatic lion is exclusively found in the dry deciduous forests of the Gir National Park in Gujarat.
- **Option D is incorrect** because while Ladakh is home to the Tibetan wild ass (Kiang), the Indian wild ass (Khur) is found specifically in the Little Rann of Kutch in Gujarat.

Q 106. Ans. B

Explanation:

- **Evaporation:** When ocean water evaporates, only the pure water turns into vapor, leaving the dissolved salts behind. Therefore, regions with high evaporation rates (like the subtropics) have **higher salinity**.
- **Precipitation:** Rain and snow add fresh, unsalty water directly to the ocean's surface, diluting it. Regions that experience heavy rainfall (like the equator) tend to have **lower salinity**.
- **River Runoff:** Rivers carry massive amounts of fresh water from the continents into the ocean. Where large rivers empty into the sea (like the Amazon or the Ganges), the surface salinity is significantly **decreased**.
- **Freezing and Melting of Sea Ice:** When seawater freezes in polar regions, the ice crystals that form are made of pure water, expelling the salt into the surrounding liquid water (a process called brine rejection). This **increases** the salinity of the unfrozen water left behind. Conversely, when that ice melts in the summer, it releases fresh water, **decreasing** the local salinity.

Q 107. Ans. A

Explanation:

- **Statement 1 is correct:** Wave height is the total vertical distance measured from the highest elevation of the wave (the crest) to the lowest depression (the trough). To understand this dynamically, oceanographers also measure the wave's amplitude, which is exactly half the wave height (the distance from the still-water line to the crest). The height of a wind-driven surface wave is dictated by three highly specific atmospheric factors: the wind's speed, the duration the wind blows, and the fetch (the uninterrupted distance of open water over which the wind blows). This is why the largest wind-driven waves on Earth consistently occur in the Southern Ocean, where the fetch is virtually infinite.
- **Statement 2 is correct:** Wavelength is the horizontal distance between two consecutive crests or two consecutive troughs. This measurement is intrinsically tied to the wave period, which is the time it takes for two successive crests to pass a single, stationary point. The wavelength determines the physical footprint of the wave's energy. Beneath the surface, the water particles move in circular orbits. The diameter of these orbits decreases rapidly with depth, and the motion stops entirely at a depth equal to exactly one-half of the wavelength, a boundary known as the wave base. If a submarine dives below this wave base, the water is perfectly calm regardless of the storm above.
- **Statement 3 is incorrect:** The speed (celerity) of a deep-water wave depends primarily on its wavelength and period, not its height. In oceanography, a wave is defined as a "deep-water wave" if the ocean depth is greater than half of its wavelength. Because the wave's energy does not reach the ocean floor, there is no friction to slow it down. Therefore, its speed is dictated strictly by gravity and wavelength ($c = \sqrt{g\lambda/2\pi}$), meaning longer waves travel faster. Height plays no structural role in this calculation. By contrast, a "shallow-water wave" (where depth is less than 1/20th of the wavelength) drags on the ocean floor, and its speed is dictated strictly by the depth of the water.

Q 108. Ans. C

Explanation:

- The Deccan Traps are one of the largest volcanic features on Earth, forming a massive "Large Igneous Province" (LIP) in west-central India.

- They were formed roughly 66 million years ago (at the Cretaceous-Paleogene boundary) when the northward-drifting Indian Plate passed directly over the **Réunion hotspot**, a stationary plume of intense heat rising from the Earth's mantle.
- This encounter triggered immense **fissure volcanism**. Highly fluid basaltic lava erupted through giant cracks in the Earth's crust, flooding vast areas and stacking up in thick, flat layers. (The term "trap" is derived from the Swedish word trappa, meaning stairs, referring to the step-like landscape these stacked lava layers created).

Q 109. Ans. D

Explanation:

- **Statement 1 is correct:** The western syntaxial bend is located near Nanga Parbat in the Kashmir region. A syntaxial bend is a sharp, hairpin-like turn in the strike of a mountain belt. At the western extremity of the Himalayas, the general east-west mountain range takes a sudden and dramatic southward plunge. This geological pivot point is visibly marked by the massive Indus River gorge, which cuts deeply through the terrain just north of Nanga Parbat before the mountain system continues south as the Sulaiman and Kirthar ranges in Pakistan.
- **Statement 2 is correct:** The eastern syntaxial bend is located near Namcha Barwa in Arunachal Pradesh. Just like the western boundary, the Himalayan range abruptly changes its alignment here, bending sharply to the south to form the Purvanchal hills of Northeast India (and extending further down into the Arakan Yoma of Myanmar). This eastern pivot is carved out by the mighty Brahmaputra River (known as the Yarlung Tsangpo in Tibet), which creates the extraordinarily deep Dihang Gorge as it wraps tightly around the Namcha Barwa peak to enter the Indian subcontinent.
- **Statement 3 is correct:** Both syntaxial bends are a direct physical result of the tectonic collision between the Indian and Eurasian plates. The Indian Plate is a massive, rigid continental block moving northward. As it crashes into the Eurasian Plate, it acts like a giant indenter. The Nanga Parbat and Namcha Barwa regions serve as the two "horns" or anchoring pivot points of this leading edge. Because the Indian Plate continues to push inward, the softer edges are pinned at these two extremities, causing the crustal rock to intensely buckle, compress, and wrap around these corners, creating the permanent hairpin bends in the mountain chain.

Q 110. Ans. B

Explanation:

- **Reykjanes Ridge:** This is the specific segment of the Mid-Atlantic Ridge that extends continuously to the south-west from the Reykjanes Peninsula of Iceland. It forms the active divergent tectonic boundary where the North American Plate and the Eurasian Plate are pulling apart. Geologically, it is highly unusual because it lacks the deep central rift valley that is typically characteristic of slow-spreading mid-ocean ridges. This anomaly is caused by the intense heat and magma upwelling from the nearby Iceland hotspot, which thickens the oceanic crust.
- **Kolbeinsey Ridge:** This is also a segment of the Mid-Atlantic Ridge connected to Iceland, but it extends to the north of the island, not the south-west.
- **Telegraphic Plateau:** As discussed in a previous geography concept, this is a relatively flat, wide section of the Mid-Atlantic Ridge located much further south, positioned between Ireland and Newfoundland. It earned its name because the first successful transatlantic telegraph cables were laid across it in the 19th century.
- **Charlie-Gibbs Fracture Zone:** This is a massive geological fault system located much further south in the Atlantic Ocean (around 52 degrees North). Instead of being a ridge, it is a system of two parallel transform faults that horizontally offset the Mid-Atlantic Ridge by hundreds of kilometres.

Q 111. Ans. C

Explanation:

- **Statement 1 is incorrect:** Admiral Krishna Swaminathan assumed charge as the 27th (not the 25th) Chief of the Naval Staff (CNS) on May 31, 2026.
 - ✓ He previously served as the Vice Chief of the Naval Staff, Flag Officer Commanding-in-Chief of the Western Naval Command, and commanded the aircraft carrier INS Vikramaditya.
- **Statement 2 is correct:** The Supreme Court of India constituted the high-powered Judicial Infrastructure Advisory Committee in May 2026 to systematically overhaul the physical and digital landscape of the Indian judiciary.
 - ✓ It aims to secure a ₹40,000 to ₹50,000 crore government allocation. A major focus of the committee is expanding the “e-courts” initiative to bridge the digital divide through robust virtual and hybrid hearing setups.

CDS	Name	Tenure Started
1st	General Bipin Rawat	January 2020
2nd	General Anil Chauhan	September 2022
3rd	General N. S. Raja Subramani	May 2026

- **Statement 3 is incorrect:** General N. S. Raja Subramani is the **third** Chief of Defence Staff (CDS) of India. He assumed the tri-services role on May 31, 2026.
 - ✓ A highly decorated infantry officer from the Garhwal Rifles, he has emphasised the strategic pillars of JAI (Jointness, Aatmanirbharta, and Integration) to accelerate military modernisation and indigenous weapons development.
- **Statement 4 is correct:** Renowned economist Ashok Kumar Lahiri was appointed as the Vice Chairperson of NITI Aayog (National Institution for Transforming India) in April 2026.
 - ✓ He replaced Suman K. Bery, who had held the government think-tank post since May 2022.
 - ✓ He previously served as the 12th Chief Economic Adviser to the Government of India (2002–2007), a full-time member of the 15th Finance Commission, the Executive Director at the Asian Development Bank, and the Chairman of Bandhan Bank.

Q 112. Ans. B

Explanation:

- **2nd WHO Global Summit on Traditional Medicine:**
 - ✓ **Theme:** Restoring Balance: The Science of Health and Well-being.
 - ✓ Hosted in New Delhi in December 2025 by the WHO and India’s Ministry of Ayush. The goal was to integrate ancient healing practices with modern science. It featured the launch of a massive global digital library to protect and study traditional medicine.
- **Global AI Show, 2025:**
 - ✓ **Theme:** AI-2031: Accelerating Intelligent Futures.
 - ✓ Held in Abu Dhabi, this major tech event focused on the next decade of Artificial Intelligence.
- **Global Energy Leaders Summit, 2025:**
 - ✓ **Theme:** Empowered India: Sufficiency, Balance, Innovation.

- ✓ **Key Facts:** Held in Puri, Odisha, this summit focused on India's transition to clean energy to meet its 2070 "net-zero" emissions target. It concluded with the "Puri Declaration," a collaborative plan for Indian states to adopt green hydrogen and resilient power grids.

● **8th International Solar Alliance Assembly:**

- ✓ **Theme:** One Sun, One World, A Shared Resolve.
- ✓ **Key Facts:** Over 120 countries met in New Delhi to expand global solar energy. India's President delivered the opening speech, a first for the assembly. They celebrated India becoming the world's third-largest solar power producer and announced new funding for solar projects in Africa.

Q 113. Ans. C

Explanation:

- **Statement 1 is correct:** The Uttar Pradesh government recently approved the establishment of India's first dedicated Forest University in Gorakhpur. Built with an estimated budget of ₹500 crore, it is being developed on 125 acres near the Jatayu Conservation and Breeding Centre. The university will serve as a national hub for specialised degrees in forestry, wildlife studies, and environmental sciences.
- **Statement 2 is correct:** INS Dhruv is India's first dedicated nuclear missile tracking and ocean surveillance ship. Built by Hindustan Shipyard Limited and operated jointly by the Indian Navy, the National Technical Research Organisation (NTRO), and the DRDO, it is a 10,000+ tonne vessel equipped with advanced AESA radars. Its primary role is to track the trajectories of ballistic missiles and gather electronic intelligence (ELINT) in the region.
- **Statement 3 is incorrect:** NPCI International Payments Limited (NIPL) signed a historic agreement with **Trinidad and Tobago's** Ministry of Digital Transformation. This made Trinidad and Tobago the first Caribbean nation to build a real-time national payments platform modelled directly on India's globally recognised UPI system.
- **Statement 4 is correct:** India's first AI-driven Human-Elephant Conflict Research Centre is being set up in the Palamu Tiger Reserve in Jharkhand. The initiative aims to shift wildlife management from traditional reactive methods to a proactive, tech-driven approach. It will use AI models to

decode elephant vocalisations (distinguishing between distress, danger, or grazing sounds) and track migratory corridors to create automated early warning systems for local villages.

Q 114. Ans. C

Explanation:

- **Statement 1 is correct:** Driven by a desire to independently maximise its oil production and monetise its heavy infrastructural investments before global oil demand plateaus, the UAE announced its departure from OPEC and OPEC+ on April 28, 2026. The exit took effect on May 1, 2026, ending a 59-year membership.
- **Statement 2 is correct:** Before its departure, the UAE was the third-largest producer in OPEC, possessing a production capacity of nearly 4.8 million barrels per day (though forced to produce much less under OPEC quota caps). Only Saudi Arabia and Iraq produced more.
- **Statement 3 is correct:** The UAE was not a founding member of OPEC. It joined the organisation in 1967 (via Abu Dhabi, before the UAE federation was officially formed in 1971).
- **Statement 4 is correct:** OPEC's Secretariat is located in Vienna, Austria. It moved there in 1965 from its original headquarters in Geneva, Switzerland. Austria has never been a member of OPEC.

Q 115. Ans. B

Explanation:

- **The Library Man of India:** The Story of P.N. Panicker:
 - ✓ **Author:** P.P. Sathyan
 - ✓ This biographical work chronicles the life of Puthuvayil Narayana Panicker, widely known as the "Father of the Library Movement" in Kerala. His grassroots campaign in the 1940s led to the establishment of thousands of rural libraries, fundamentally driving Kerala toward achieving 100% literacy. In his honour, the Government of India observes June 19 (his death anniversary) as National Reading Day.
- **All-Time Favourite Friendship Stories:**
 - ✓ **Author:** Ruskin Bond
 - ✓ Released by the beloved Mussoorie-based author, this anthology brings together his classic, heartwarming tales of camaraderie, human connection, and the innocence of youth in the Himalayan foothills. Bond, a recipient of both the Padma Shri and Padma

Bhushan, continues to be one of India's most prolific writers for children and young adults.

- **Better Never Stops: Leadership Insights from Life's Unscripted Moments:**

- ✓ **Author:** Shyam Srinivasan
- ✓ Authored by the veteran banker and long-serving former Managing Director & CEO of Federal Bank. The book acts as a memoir and a leadership manual, distilling his decades of corporate experience. He discusses how navigating unforeseen challenges—the “unscripted moments”—is what truly shapes successful leadership and institutional resilience.

- **The Sage Who Reimagined Hinduism:**

- ✓ **Actual Author:** Dr. Shashi Tharoor
- ✓ **The full title of the book is The Sage Who Reimagined Hinduism:** The Life, Lessons and Legacy of Sree Narayana Guru. It explores the profound impact of Sree Narayana Guru, the revered 19th-century social reformer from Kerala who fought against the rigid caste system and propagated the universal message of “One Caste, One Religion, One God for All” (Oru Jathi, Oru Matham, Oru Daivam, Manushyanu).

Q 116. Ans. B

Explanation:

- **Assertion (A) is correct:**

- ✓ India's Prototype Fast Breeder Reactor (PFBR) is located at Kalpakkam, Tamil Nadu. It was designed by the Indira Gandhi Centre for Atomic Research (IGCAR) and built by BHAVINI (Bharatiya Nabhikiya Vidyut Nigam Limited).
 - The PFBR successfully achieved its “first criticality” (a self-sustaining nuclear chain reaction). This historic milestone marks India's official entry into the second stage of its three-stage nuclear power program.

- **Reason (R) is correct:**

- ✓ The defining characteristic of a “breeder” reactor is that it generates (breeds) more fissile material than it consumes.
- ✓ The PFBR uses a Mixed Oxide (MOX) fuel made of Plutonium and Uranium. The reactor core is surrounded by a “blanket” of fertile Uranium-238. When fast neutrons from the core strike this blanket, the U-238 captures a neutron and transmutes into Plutonium-239 (a fissile material). It also uses liquid sodium as a coolant rather than water, which allows the neutrons to remain “fast” (unmoderated).

- **Why (R) does not explain (A):**

- ✓ While both statements are scientifically and factually accurate, the fundamental physics of how a breeder reactor operates (Reason R) does not explain its geographical placement in Tamil Nadu (Assertion A). The Kalpakkam site was chosen for the PFBR because it already houses the Madras Atomic Power Station (MAPS), is located near the ocean for a massive supply of cooling water, and is the home of IGCAR, which developed the reactor's technology.

Q 117. Ans. B

Explanation:

- **Asian Boxing Championships → 2 (Gold, Vishwanath Suresh):** During the U-19 and U-23 Asian Boxing Championships held in Jakarta, Indonesia in **2026**, Asian champion Vishwanath Suresh led the Indian contingent's dominant campaign, competing and securing top honours in the U-23 men's 50kg category.
- **Payal Nag → 1 (Gold, World Para Archery):** On **April 4, 2026**, 18-year-old quadruple amputee Payal Nag created history at the World Archery Para Series in Bangkok. In only her second international event, she defeated her idol, World No. 1 and Paralympic star Sheetal Devi, with a score of 139–136 to clinch the gold medal in the Compound event.
- **IBSF World Billiards Championship → 3 (Sourav Kothari beat Pankaj Advani):** On **April 29, 2026**, Sourav Kothari successfully defended his IBSF World Billiards Championship title. He outclassed 19-time champion and compatriot Pankaj Advani with a massive scoreline of 1133-477 in a dominant three-hour final.
- **National Senior Federation Cup → 4 (Samardeep Singh Gill):** On **May 25, 2026**, at the 29th National Senior Athletics Federation Competition (Federation Cup) in Ranchi, 25-year-old Samardeep Singh Gill caused a major upset. He defeated two-time Asian Games champion and national record holder Tajinderpal Singh Toor to win gold in the men's shot put, throwing a distance of 20.46m to successfully breach the 2026 Commonwealth Games qualification mark.

Q 118. Ans. B

Explanation:

- **Bhu-Bharati Portal → Telangana (2):** Rolled out in **2026**, this is the state's newly integrated digital land record management system. It was specifically launched to overhaul and replace the highly criticised

'Dharani' portal, streamlining property registrations, mutation checks, and encumbrance certificates.

- **Majhe Gaon Arogya Sampanna Gaon → Maharashtra (4):** Officially launched on **April 1, 2026** (with an ₹80.75 crore budget). Translating to "My Village, A Healthy Village," this is a massive rural initiative focusing on preventive healthcare, sanitation, and disease control. Villages that score above 70% on health parameters receive cash prizes of up to ₹5 crore.
- **E-Shram Saathi App → Chhattisgarh (1):** Launched on **May 1, 2026** (International Labour Day) by CM Vishnu Deo Sai. It functions as a "digital labour chowk," directly connecting unorganised workers with employment opportunities and centralised state welfare schemes.
- **KhuaReL App → Mizoram (3):** Launched on **April 29, 2026** by the state's Disaster Management & Rehabilitation (DM&R) department. "KhuaReL" translates roughly to nature/natural occurrences in Mizo. It is a real-time incident reporting system, making Mizoram the first Northeastern state to deploy such an advanced digital disaster management application.

Q 119. Ans. D

Explanation:

- **Exercise Tiger Triumph (India–USA):** This is a massive, bilateral tri-service amphibious military exercise. The most recent 4th edition took place from **April 1 to April 13, 2025**. The harbour phase was conducted in Visakhapatnam, and the sea phase, which included massive amphibious landing exercises, took place in Kakinada on the Eastern Seaboard. Notably, the United States Space Force participated with their Indian counterparts for the first time during this edition.
- **Exercise Varuna (India–France):** This is a high-profile, annual bilateral naval exercise that forms a cornerstone of the India-France strategic partnership. The recent 23rd edition was held from **March 19 to March 22, 2025**. It featured complex multi-domain operations, including advanced air defence drills involving Rafale-M jets from the French Navy and MiG-29K jets from the Indian Navy, alongside anti-submarine warfare exercises.
- **Exercise SIMBEX (India–Singapore):** The Singapore-India Maritime Bilateral Exercise (SIMBEX) is the longest uninterrupted naval exercise India has with any other country (running since 1994). The 32nd edition was hosted by

Singapore from **July 28 to August 1, 2025**. The Indian Navy was prominently represented by INS Satpura, participating in advanced warfare drills and maritime security operations in the southern parts of the South China Sea.

Q 120. Ans. C

Explanation:

- **Madhya Pradesh** became the first Indian state to implement an Artificial Intelligence (AI)-based Real-Time Forest Alert System.
- The pilot project was officially launched in **May 2025**.
- The system operates using the Google Earth Engine, which analyses multi-temporal satellite imagery through a custom-built machine learning model. It detects illegal tree felling, forest land encroachments, and land-use changes in areas as small as 10×10 meters. Alerts are sent directly to the mobile apps of field staff every 2-3 days, equipped with GPS coordinates, making it vastly superior to older manual reporting systems.
- **It was initially tested in five sensitive forest divisions known for high encroachment:** Shivpuri, Guna, Vidisha, Burhanpur, and Khandwa.

Q 121. Ans. A

Explanation:

- **PM-KISAN → 1 (Ministry of Agriculture & Farmers Welfare):**
 - ✓ **Launch Date:** Officially launched on **February 24, 2019** (retrospectively effective from December 1, 2018) by PM Narendra Modi in Gorakhpur, Uttar Pradesh.
 - ✓ It is a Central Sector Scheme (100% funded by the Government of India) that provides an income support of ₹6,000 per year to all landholding farmer families across the country. This amount is transferred directly into their bank accounts in three equal instalments of ₹2,000 every four months.
- **PMJDY → 2 (Ministry of Finance):**
 - ✓ **Launch Date:** Launched on **August 28, 2014**.
 - ✓ The Pradhan Mantri Jan Dhan Yojana is India's National Mission for Financial Inclusion. It ensures access to financial services like savings & deposit accounts, remittance, credit, insurance, and pensions in an affordable manner. Key benefits include a zero-balance account facility, an inbuilt RuPay debit card with an accidental insurance cover of ₹2 lakh, and an overdraft facility up to ₹10,000.

● **e-Shram Portal → 3 (Ministry of Labour & Employment):**

- ✓ **Launch Date:** Launched on **August 26, 2021**.
- ✓ This is the first-ever national database constructed to register unorganised workers (including construction workers, migrant labourers, gig workers, street vendors, and domestic workers). Registered workers receive a unique 12-digit **Universal Account Number (UAN) card**, which seamlessly links them to various social security benefits and accidental insurance policies administered by the Central and State governments.

● **India Semiconductor Mission (ISM) → 4 (Ministry of Electronics & Information Technology):**

- ✓ **Launch Date:** Setup in **December 2021**.
- ✓ ISM was established as a specialised, independent business division within the Digital India Corporation to drive India's long-term strategies for building a sustainable semiconductor and display ecosystem. It handles the structural implementation of the ₹76,000-crore incentive scheme aimed at attracting global semiconductor manufacturing hubs and fabrication plants to India.

Q 122. Ans. C

Explanation:

- **Statement 1 is correct:** In a historic 2026 legal first, Spain's Mar Menor Lagoon exercised its legal personhood by actively suing an agricultural enterprise in court over illegal wastewater pollution.
- **Statement 2 is correct:** New Zealand passed a legislative framework recognising Mount Taranaki as a legal person. Crucially, the mountain is not state property; it legally owns itself and is governed by a joint council.
- **Statement 3 is correct:** A municipality in the Peruvian Amazon passed a binding ordinance recognising native stingless bees as legal subjects, allowing environmentalists to sue over habitat destruction.
- **Statement 4 is incorrect:** The Los Cedros cloud forest is located in **Ecuador**, not Costa Rica. Ecuador is famous for enshrining the Rights of Nature directly into its national Constitution, which the forest is now using to claim copyright royalties to fund its own conservation.

Q 123. Ans. D

Explanation:

- **Statement 1 is correct:** The Global Biofuels Alliance

(GBA) was officially launched on **September 9, 2023**, on the sidelines of the G20 Summit held in New Delhi during India's G20 Presidency.

- **Statement 2 is correct:** India is a core founding member. The initiative was spearheaded by **India, the United States, and Brazil**, which together account for a massive majority of global ethanol production and consumption. Other founding members included Argentina, Canada, Italy, and South Africa.
- **Statement 3 is correct:** The alliance's primary objective is to facilitate international cooperation, intensify the use of sustainable biofuels, and develop global standards and certification systems to help accelerate the global transition to clean energy.

Q 124. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The second edition of this major bilateral event was held in **Lumbini, Nepal**. The festival specifically celebrated the shared Buddhist heritage of the two nations, featuring cultural performances (like Kathak and traditional Nepali dance) and academic seminars on cross-border linkages.
- **Pair 2 is correctly matched:** The 6th Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) Summit was hosted by **Bangkok, Thailand**. The summit operated under the theme "Prosperous, Resilient, and Open BIMSTEC," and featured India announcing several key initiatives, including plans to host the first BIMSTEC Games in 2027.
- **Pair 3 is correctly matched:** The 2025 Shanghai Cooperation Organisation (SCO) Heads of State Council meeting was hosted by **China** in the city of **Tianjin** from **August 31 to September 1, 2025**. This marked the fifth time China hosted the event, acting under the theme "Upholding the Shanghai Spirit: SCO on the Move."
- **Pair 4 is incorrectly matched:** The inaugural World Audio Visual & Entertainment Summit (WAVES) was hosted in **Mumbai, Maharashtra** (at the Jio World Convention Centre), not Delhi. The landmark four-day summit took place from **May 1 to May 4, 2025**, drawing delegates from over 90 countries to position India as a global hub for the media, animation, and entertainment economy.

Q 125. Ans. C

Explanation:

- **International Year of Glaciers' Preservation → 2 (2025):** The UN General Assembly adopted

a resolution (spearheaded by Tajikistan) to declare 2025 as the International Year of Glaciers' Preservation. It aims to raise global awareness about the accelerated melting of glaciers and their severe impact on the global water cycle and sea levels.

- **International Year of Millets → 1 (2023):** Spearheaded by the Government of India and supported by 72 countries, the UN declared 2023 as the International Year of Millets. The goal was to position India as the "Global Hub for Millets" (branded domestically as Shree Anna) and highlight their climate resilience and nutritional superiority.
- **International Year of Camelids → 3 (2024):** The UN designated 2024 to highlight the crucial economic and cultural importance of camelids (which include camels, llamas, alpacas, and vicuñas). They are vital for the food security and livelihoods of millions of families residing in some of the most hostile environments on Earth, spanning from the Andean highlands to the deserts of Africa and Asia.
- **International Year of Quantum Science and Technology → 4 (2025):** In June 2024, the United Nations officially proclaimed 2025 as the International Year of Quantum Science and Technology. This specific year was chosen to mark exactly 100 years since the initial foundational development of quantum mechanics by Werner Heisenberg and others in 1925.

Q 126. Ans. C

Explanation:

- **Statement 1 is correct:** The Global Liveability Index is an annual flagship report published by the **Economist Intelligence Unit (EIU)**, which is the research and analysis division of The Economist Group.
- **Statement 2 is correct:** The EIU evaluates and ranks 173 cities globally based on 30 qualitative and quantitative factors grouped into five broad categories: **Stability** (25%), **Healthcare** (20%), **Culture and Environment** (25%), **Education** (10%), and **Infrastructure** (20%).
- **Additional Facts:**
 - ✓ In the **June 2025** release, **Copenhagen (Denmark)** overtook Vienna (Austria) to become the world's most liveable city. Vienna had previously held the rank 1 spot for three consecutive years but dropped to rank 2 due to a slight decline in its stability score.
 - ✓ **The top of the 2025 index is dominated by Western Europe and Australia:** Copenhagen

(1st), Vienna and Zurich (tied 2nd), Melbourne (4th), and Geneva (5th).

- ✓ Damascus (Syria) continues to remain at the absolute bottom (173rd) of the index due to ongoing geopolitical conflict and devastated infrastructure.
- ✓ No Indian city made the top tier. Developing cities typically struggle in this index primarily due to low scores in the Infrastructure (traffic, housing, water provision) and Environment parameters.

Q 127. Ans. D

Explanation:

- **Pair 1 is correctly matched:** In **March 2025**, the Reserve Bank of India (RBI) granted a Non-Banking Financial Company (NBFC) license to Flipkart Finance Private Limited. This made Flipkart the first major e-commerce platform in India to secure regulatory approval for direct lending. It allows the Walmart-backed company to offer credit (like personal loans and BNPL) directly to customers and sellers via its super. money app without relying exclusively on third-party banks. However, as an NBFC, it is not permitted to accept public deposits.
- **Pair 2 is correctly matched:** On **April 21, 2026**, the Indian Institute of Technology (IIT) Ropar launched **ANNAM.AI**, recognised as India's first fully integrated agricultural intelligence ecosystem. Backed by the government and global tech partners, its three-layer architecture consists of: Swan (advanced micro-climate weather stations), Krishi AI (computer vision for pest detection), and ACE (Annam Chat Engine for multilingual, expert-validated advisories).
- **Pair 3 is correctly matched:** In **December 2025**, the Ministry of Electronics and Information Technology (MeitY) officially launched **DHRUV-64**, India's first homegrown 1.0 GHz, 64-bit dual-core microprocessor. It was developed by C-DAC under the Digital India RISC-V (DIR-V) program. Based on open-source, royalty-free RISC-V architecture, it eliminates foreign licensing fees and is designed to secure critical sectors like 5G infrastructure, defence, and IoT.

Q 128. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The Nagaland government recently issued a strict directive prohibiting the display of religion-based,

NGO-based, and other unauthorised slogans or signage on both private and public vehicles. While the state claimed it was to enforce the Motor Vehicles Act, it sparked significant pushback from local trade and church bodies who argued that such community identifiers are vital during emergencies in the state's unique socio-political environment.

- **Pair 2 is correctly matched:** In May, the Karnataka government officially revoked the controversial February 2022 government order that had effectively barred the hijab in educational institutions with prescribed uniforms. The state's Department of School Education and Literacy issued new guidelines permitting students to wear limited traditional and religious symbols—including the hijab, turban, and sacred thread, provided they do not interfere with discipline or student identification.
- **Pair 3 is incorrectly matched:** On July 1, 2026, the Uttarakhand government officially dissolved the State Madrasa Board as the new Uttarakhand Minority Education Act came into force. The state established a new State Minority Education Authority to replace the board. Under this new framework, madrasas and minority schools are required to seek recognition from the education department, follow the standard NCERT curriculum, and integrate students into mainstream education while continuing religious instruction.
- **Pair 4 is correctly matched:** On April 27, 2026, the Meghalaya government promulgated the Meghalaya Official Languages Ordinance, 2026. This historic move officially elevated both **Khasi** and **Garo** to the status of official languages of the state alongside English. Previously, under the 2005 State Language Act, they only held "associate" status despite being spoken by the overwhelming majority of the state's indigenous population.

Q 129. Ans. B

Explanation:

- Puri's Golden Beach received Blue Flag certification for the **seventh consecutive year** (2025-26), and Sunapur Beach in Ganjam district received it for the **third consecutive year** (2025-26), confirmed by the Foundation for Environmental Education (FEE), Denmark, in November 2025.

Q 130. Ans. A

Explanation:

- **Armenia** officially became the newest member of the International Union for Conservation of Nature

(IUCN). This was announced during the IUCN World Conservation Congress held in **Abu Dhabi** in October 2025.

- Armenia is actively stepping up its global environmental leadership, as the country is preparing to host the 17th Conference of the Parties to the Convention on Biological Diversity (CBD COP17) in 2026.
- Situated at the crossroads of Europe and Asia, Armenia is home to diverse ecosystems and critically endangered species, such as the Caucasian leopard and the endemic Sevan trout. By joining the IUCN, Armenia gains access to global expertise and international financing to help achieve its pledge of restoring 12.9% of its forest territory by 2030.

Q 131. Ans. D

Explanation:

- **One Battle After Another** → 4 (**BAFTA Best Film**): Paul Thomas Anderson's politically charged 2026 thriller dominated the British Academy Film Awards, taking home Best Picture alongside directing and acting accolades.
- **It Was Just an Accident** → 3 (**Palme d'Or**): Iranian director Jafar Panahi's 2025 black comedy thriller had its world premiere at the 78th Cannes Film Festival, where it won the prestigious Palme d'Or.
- **12th Fail** → 2 (**National Film Award – Best Film**): Vidhu Vinod Chopra's universally acclaimed biographical drama was honoured with the Best Feature Film award at the 71st National Film Awards.
- **Hamnet** → 1 (**BAFTA Best Actress**): Jessie Buckley swept the 2026 awards season, including winning the BAFTA for Best Leading Actress, for her moving portrayal of Agnes Hathaway in this Shakespearean family tragedy.

Q 132. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Authored by former journalist and lawyer **Berjis Desai**. The book is a political biography and analysis of Prime Minister Narendra Modi, focusing on his ideological roots in the Sangh and his governance model.
- **Pair 2 is correctly matched:** Authored by **Ashok Tandon**, a veteran journalist who served as the media advisor to former Prime Minister Atal Bihari Vajpayee. The book (originally released in Hindi as *Atal Sansmaran*) chronicles behind-the-scenes political memories and foreign policy decisions during the Vajpayee era.

- **Pair 3 is correctly matched:** Published in 2025, this book contributes to the expanding field of civilisational studies and post-colonial identity politics in India. **Abhijit Joag** engages in a critical re-examination of Indian historiography, aiming to deconstruct colonial-era narratives and advocate for a revitalisation of indigenous conceptual frameworks within the nation's contemporary socio-political identity.
- **Pair 4 is correctly matched:** Authored by **Sudha Murty**. It is the latest children's adventure novel featuring her beloved recurring character, Nooni.

Q 133. Ans. B

Explanation:

- The **Tianwen-2** mission is an ambitious deep-space exploration project conducted by the **China National Space Administration (CNSA)**.
- The mission was launched on **May 28, 2025**, from the Xichang Satellite Launch Center via a Long March 3B launch vehicle.

Q 134. Ans. A

Explanation:

- In June 2025, the **India Post Payments Bank (IPPB)** was officially conferred the prestigious **Digital Payments Award 2024–25** by the Department of Financial Services, Ministry of Finance.
- The award recognised IPPB for its outstanding contribution to financial inclusion and its role in expanding the digital payment ecosystem, particularly in rural and underserved areas.
- IPPB achieved the **1st position** among Payments Banks in India in the performance index for the financial year 2024–25.
- The award was presented by the Union Minister of Finance and Corporate Affairs, Smt. Nirmala Sitharaman, to the leadership team of IPPB at a ceremony held in New Delhi.

Q 135. Ans. B

Explanation:

- Elon Musk became the world's first trillionaire on **June 12, 2026**, when SpaceX began trading on the Nasdaq at \$150 per share under the ticker SPCX.
- His combined stake in SpaceX and Tesla pushed his net worth to an estimated \$1.1 trillion, as declared by Forbes.
- The SpaceX IPO raised \$75 billion, the largest IPO in history, surpassing Saudi Aramco's 2019 record.

Q 136. Ans. B

Explanation:

- **Rithika Sri** from **Tamil Nadu** became India's first transgender cricket umpire in April 2026. The Tamil Nadu Cricket Association enabled this milestone by introducing an "Other" gender category in its State Panel Umpire examination.
- She began her umpiring career in 2021 and has officiated matches across Salem, Namakkal, and Coimbatore districts.

Q 137. Ans. C

Explanation:

- **Madrid Open (Women's Singles):** Ukrainian tennis player **Marta Kostyuk** secured her first WTA 1000 title at the Madrid Open 2026, defeating Mirra Andrejeva in the final.
- **Madrid Open (Men's Singles):** World No. 1 **Jannik Sinner** continued his dominant 2026 form, defeating Alexander Zverev to win the Madrid Open men's singles title.
- **La Liga: Barcelona** secured the 2025–26 La Liga title (the 95th edition of the competition), clinching their second consecutive championship with a victory over Real Madrid.
- **English Premier League: Arsenal** was crowned the 2025/26 English Premier League champions, ending a 22-year title drought.

Q 138. Ans. B

Explanation:

- **Statement 1 is correct:** According to the Economic Survey 2025–26, the services sector's share in India's GDP rose to **53.6%** in the first half of FY26 (H1 FY26), reflecting the sector's continued dominance as a growth engine for the Indian economy.
- **Statement 2 is incorrect:** As per the Economic Survey 2025–26, India is ranked as the world's **seventh-largest** exporter of services, not the fifth. The Survey highlights that India's share in global services trade has more than doubled from 2% in 2005 to 4.3% in 2024, cementing its position among the top global services exporters, but it has not yet reached the 5th position.

Q 139. Ans. A

Explanation:

- **Pair 1 is correctly matched:** The **Indira Gandhi Prize for Peace, Disarmament and Development for 2025** was awarded to the Mozambican rights

activist and humanitarian **Graça Machel**. The award recognises her extensive and tireless advocacy for peace, social development, and the protection of the rights of women and children globally.

- **Pair 2 is incorrectly matched:** The **2026 Abel Prize** was awarded to the German mathematician **Gerd Faltings**, of the Max Planck Institute for Mathematics, for his fundamental contributions to arithmetic geometry. The mathematician Masaki Kashiwara was actually the recipient of the 2025 Abel Prize.
- **Pair 3 is incorrectly matched:** The **2026 World Food Prize** was awarded to the Dutch scientist **Huub Lelieveld**. He is recognised as the “Architect of Global Food Safety Standards” for his work in establishing the Global Harmonisation Initiative (GHI) and his global efforts to improve food safety and reduce food loss. Mariangela Hungria is a renowned Brazilian researcher in agriculture, but she is not the 2026 World Food Prize laureate.

Q 140. Ans. C

Explanation:

- **Statement 1 is correct:** The PM-VBRY (originally announced as the Employment Linked Incentive Scheme in the Union Budget 2024-25) officially came into effect on **August 1, 2025**, covering new jobs created up to July 31, 2027. Prime Minister Narendra Modi formally launched the scheme’s first major disbursement of incentives (worth ₹2,400 crore) on **June 19, 2026**.
- **Statement 2 is incorrect:** The scheme is administered by the **Ministry of Labour and Employment**, and is operationally implemented through the Employees’ Provident Fund Organisation (EPFO), not the Ministry of Skill Development and Entrepreneurship.
- **Statement 3 is correct:** The scheme targets formalisation of the workforce. Absolute first-time EPFO-registered employees earning a gross monthly wage of up to **₹1 lakh** are eligible to receive a direct financial incentive (equivalent to one month’s EPF wage) up to a maximum of **₹15,000**, paid in two instalments via Direct Benefit Transfer (DBT).

Q 141. Ans. D

Explanation:

- **Given statements:** $B > A \geq C \leq G < D$; $E = F > G > H \geq J$
- **On combining, we get:**
 - ✓ $B > A \geq C \leq G > H \geq J$; $B > A \geq C \leq G < F = E$; $E = F > G < D$; $D > G > H \geq J$

● Conclusions:

- ✓ $E < A$: False (As, $A \geq C \leq G < F = E$, so, no direct relation between E and A can be established)
- ✓ $D > J$: True (As, $D > G > H \geq J$, so, $D > J$)
- ✓ $C < F$: True (As, $C \leq G < F$, so, $C < F$)

● Therefore, only conclusions II and III are true.

Q 142. Ans. D

Explanation:

● Given word:

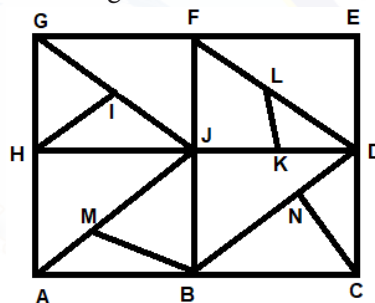
● ‘METROPOLITAN’

- OP, OT, PT, LO, OP, OR and PR are 7 such pairs which have as many letters between them (both forward and backward) as in English alphabetical series.

Q 143. Ans. A

Explanation:

Given figure:

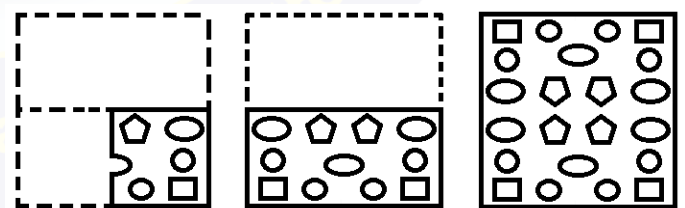


Triangles are: GHI, HIJ, HGJ, GJF, KDL, DJF, DEF, ABM, MJB, ABJ, AJH, BNC, CND, BCD, BJD, AJG and BDF.

So, there are 17 triangles in the given figure.

Q 144. Ans. C

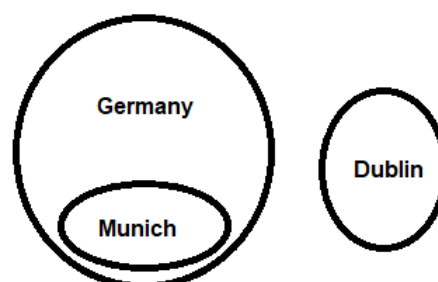
Explanation:



Q 145. Ans. C

Explanation:

The relationship is shown below.



Munich is a city in Germany. But Dublin is a city in Ireland.

Q 146. Ans. B

Explanation:

- Let the original number be x .
- According to the problem:
 - ✓ $3(x + 12) = 54$
 - ✓ $\Rightarrow x + 12 = 18$
 - ✓ $\Rightarrow x = 6$
- Therefore, the original number is 6.

Q 147. Ans. B

Explanation:

- Speed of the train = 72 km/h
 - ✓ $= 72 \times 5/18 = 20$ m/s
- Total distance = Speed $\times$ Time
 - ✓ $= 20 \times 60$
 - ✓ $= 1200$ meters
- Length of the bridge = Total distance - Length of the train
 - ✓ $= 1200 - 900$
 - ✓ $= 300$ meters

Q 148. Ans. D

Explanation:

- Let the number = N
 - ✓ $\therefore \text{Remainder} \left\{ \frac{N}{8} \right\} = 5$
- Least value of $N = 5$
 - ✓ $\therefore \text{Remainder} \left\{ \frac{N^2}{8} \right\} = \left\{ \frac{(5)^2}{7} \right\}$
 - ✓ $\therefore \text{Remainder} \left\{ \frac{25}{8} \right\} = 1$

Q 149. Ans. B

Explanation:

- Daily production = 2,400 shirts
- Defective shirts = 5% of 2,400 = 120 shirts
- Effective (usable) shirts produced per day = 2,400 - 120 = 2,280 shirts
- Number of days required = $68,400 \div 2,280 = 30$ days
- Therefore, the correct answer is: B. 30 days.

Q 150. Ans. A

Explanation:

- Let the cost price be x .
- Profit at selling price 3600 = $3600 - x$
- Profit at selling price 2400 = $2400 - x$
- Given:
 - ✓ $3600 - x = 3(2400 - x)$
 - ✓ $3600 - x = 7200 - 3x$
 - ✓ $2x = 3600$
 - ✓ $x = 1800$
 - ✓ Cost price = 1800
- For a 25% loss:
 - ✓ Selling price = 75% of 1800
 - ✓ $= (75/100) \times 1800$
 - ✓ $= 1350$
- The correct selling price is 1350. Since 1350 is not in the options, the intended answer is most likely 1400 due to an error in the question or options.

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